Syntax proposes, prosody decides: On Lithuanian *si*

Víctor Acedo-Matellán
Veronika Gvozdovaitė

Abstract

We develop a syntactic-prosodic account of the linearization of the Lithuanian reflexive *si*, which may appear as a prefix or a suffix. Syntax linearizes *si* as left-adjacent to the verb. If no other prefix is present, however, the prosodic component ensures *si* ends up suffixed to the base, so that it avoids the left edge of a maximal prosodic word. The variable linearization of *si* falls together with that of other prosodically sensitive elements crosslinguistically. We show the empirical inadequacy of previous morphological and syntactic accounts, and we explore predictions related to morphological cycles and segmental phonology.

Keywords: Lithuanian *si*, Mirror Theory, Optimality Theory, prefixes, Spanning, syntax-prosody interface

1 Introduction

The Lithuanian reflexive *si* can appear either as a suffix (1a) or as a prefix (1b).

(1) a. mato-**si**
see.PRS.3-*si*
'is/are visible'

b. ne-**si**-mato
not-*si*-see.PRS.3
'is/are not visible'

In most of the generative literature, the linearization of *si* is described as a second-position phenomenon, and is explained either via syntactic movement or post-syntactic morphological operations (Embick and Noyer 2001, Embick 2007, Arregi and Nevins 2012, Korostenskienė 2014, 2016a, 2016b, Šereikaitė 2017, Adamson 2019, Bruening 2019).[1] We will show that syntactic and morphological accounts are not able to capture the full range of the behavior of *si*, as attested in the received body of data, and make the wrong predictions regarding new evidence that we contribute here.

Based on morphosyntactic and semantic evidence, we analyze *si* as the exponent of a Voice head (see also Kushnir 2025). Assuming the Spanning framework to syntax and its interfaces with the lexicon, morphology, and phonology (Bye and Svenonius 2012, Svenonius 2016, Acedo-Matellán and Kwapiszewski 2026), we propose that syntax linearizes *si* as a prefix adjacent to the verbal base. When the morphosyntactic output reaches the prosodic level, an Optimality-Theoretic algorithm selects a final representation in which *si* is not aligned with the left boundary of the maximal projection of a prosodic word. This has the effect of displacing *si* to a suffixal position in cases in which the verb shows no other prefix. Crucially, this displacement is insensitive to any morpheme boundaries: when a suffix, *si* appears as the rightmost exponent of the word, to the right of overt verbalizers and tense and agreement inflection in verbs, and to the right of nominalizers and adjectivizers and associated inflectional

morphology in nominalizations and adjectival participles.

Such prosodic explanation is supported crosslinguistically, *si* behaving like edge-sensitive elements in other languages. To illustrate, we draw a comparison between the behavior of *si* and that of pronominal clitics in European Portuguese, like *se* in (2), which have a similar distribution and for which an analogous prosodic constraint has been demonstrated (Barbosa 1996).

(2) a. vê-**se**
see.PRS.3SG-REFL.3
'is visible'

b. não **se** vê
not REFL.3 see.PRS.3SG
'is not visible'

We show that the two phenomena, in Lithuanian and in Portuguese, are of the same nature, only sensitive to different levels of the prosodic hierarchy. This is a second-position phenomenon, but, crucially, it is determined prosodically, and not morphologically or (wholly) syntactically, contra previous proposals. Our analysis thus does away with postsyntactic operations like Local Dislocation (Embick and Noyer 2001), already put into question by Korostenskienė (2017) and Šereikaitė (2017, 2022). Furthermore, it shows that, regarding linearization, syntax determines a default state of affairs, while prosody has the final say.

Our prosody-based account makes certain predictions concerning both morphological and phonological facts. As regards the former, we predict that *si* will appear as a suffix even if the verbal base is extended with further derivational processes, like nominalizations, as the position of *si* is insensitive to morphology. As regards the latter, and contra certain morphological accounts like Stump's (2000), segmental effects reveal that *si* behaves as a suffix when appearing after the base, and as a prefix when appearing before the base. We show that these predictions are borne out.

Section 2 presents the received description of the distribution of *si* and shows that

previous accounts are empirically inadequate. We present our empirical generalizations in section 3. Section 4 lays out the theoretical assumptions in our account. In section 5, we derive the default linearization of *si* as prefix, dictated by syntax. In section 6 we develop an analysis whereby the final position of *si* is prosodically determined. Section 7 explores the predictions of our proposal. Section 8 concludes.

2 A Critical State of the Art

2.1 The Received Description of the Distribution of si

Most research on the linear placement of *si* descriptively situates it among a class of elements that occur in second position: a Wackernagel clitic of sorts (Wackernagel 1892). Nevis and Joseph (1993) are, to our knowledge, the first authors to suggest this, and many later accounts are based on their description of *si*. Embick and Noyer (2001), Billings (2002), Arregi and Nevins (2012), Adamson (2019), and Bruening (2019) all propose a flavor of the second-position analysis.

Unfortunately, this strand of research has used a flawed description of the distribution of *si* in Nevis and Joseph 1993. These authors correctly describe *si* as a prefix appearing between a single other prefix and the verbal base (1b), and as a suffix to the verb when there is no other prefix (1a). However, they also argue that, in forms showing more than one prefix, *si* always occurs after the first prefix (Nevis and Joseph 1993:96). This is straightforwardly falsified by examples like (3), which shows that *si* is always the last prefix, regardless of the nature of the other prefixes: either superlexical prefixes like aspectual *be-*, or lexical prefixes (LPs) like *pa-* (Svenonius 2004, Korostenskienė 2015, 2017).[2]

(3) Te-ne-be-pa-**si**-mato.
te-NEG-PROG-LP-*si*-see.PRS.3
'Let them not see each other again.'

The description in Nevis and Joseph 1993 is thus not representative of the distribution of *si*, in spite of which most accounts have been based on it, making the wrong predictions. To our knowledge, only the following authors state the linearization of *si* in relation to inflected verbs correctly: Korostenskienė (2014, 2016a, 2016b, 2017), Šereikaitė (2017), Stump (2022), Kushnir (2025), and, in unpublished work, Bruening (2019) and Šereikaitė (2022). However, their accounts are inadequate in other respects, as we will show.

2.2 Existing Analyses of the Placement of si

There are two main families of analyses of *si* placement: it is either approached as a linear (morphological) phenomenon or as a hierarchical (i.e. syntactic) one. Embick and Noyer (2001), Embick (2007), Arregi and Nevins (2012), Šereikaitė (2017), and Adamson (2019) all argue for a post-syntactic account, where *si* is displaced rightwards in an already linearized representation. Stump 2022 is also a non-hierarchical account, although developed within a Word Formation Rules framework. Other accounts, like Bruening 2019 and Korostenskienė 2014, 2016a, 2016b, argue that the positioning of *si* operates on a hierarchical, syntactic representation. We examine these two kinds of approaches in the next two subsections, leaving Kushnir's (2025) approach, close to ours, for section 6.2.

2.2.1 Linear Accounts

The accounts in the Distributed Morphology (DM) literature (Embick and Noyer 2001, Embick 2007, Arregi and Nevins 2012, Adamson 2019, Šereikaitė 2017) use post-syntactic operations to derive the linear position of *si*. We illustrate with Embick and Noyer's (2001) account.

These authors take *si* to be a dissociated morpheme, added post-syntactically on the way to PF, and therefore not impacting on LF. This is the first problem with their analysis, since, as will be argued in section 3.2, there is evidence that *si* is not syntactico-semantically inert. They take *si* to be initially linearized to the left of the linearized syntactic object, with or without prefixes. In the representation below $*$ indicates adjacency between *subwords*, which are morphological units under the level of the *morphological word* (see Embick 2010:37).

(4) [si $*$ [LP $*$ V $*$ T]] (e.g. [si [* pa * mat * o]] 'is/are visible')

Si then trades left-adjacency for right-adjunction to the left-peripheral subword via an operation of Local Dislocation (LD), a post-syntactic, post-Vocabulary Insertion operation that operates on subwords ($\bigoplus$ indicates that two adjacent sequences form a single subword).

(5) [LP $\bigoplus$ si $*$ ... $*$ T]

However, and as observed by Korostenskienė (2017) and Šereikaitė (2017, 2022), the account crashes when a derivation contains multiple prefixes: it predicts that *si* will appear as a second prefix, contrary to fact. We illustrate this in (6).

(6) a. *Prediction:* [si $*$ [Neg $*$ LP $*$ V $*$ T]] $\rightarrow$ [Neg $\bigoplus$ **si** $*$ LP $*$ V $*$ T]

b. *Data:* ne-pa-**si**-mat-o, ne-be-pa-**si**-mat-o, *ne-**si**-pa-mat-o

To this linearization problem, it must be added that this account makes the wrong prediction that *si* will behave phonologically as a suffix with respect to the prefix with which it forms a subword (see section 7.2).

Moreover, the derivation of unprefixed forms is also inviable. Embick and Noyer (2001) observe that their LD procedure would generate V-*si*-T sequences in prefixless cases, contrary to fact, as evidenced by the ungrammaticality of (7).

(7) *mat-**si**-o
see-*si*-PRS.3
Int.: 'is/are visible'

To avoid this, Embick and Noyer (2001) have to propose that T in Lithuanian always undergoes string-vacuous Local Dislocation, creating one single subword out of the two adjacent subwords V and T. This new subword, as such, is impenetrable to LD.

(8) $[\ \text{si} * [\ \text{V} \oplus \text{T}\ ]\ ] \rightarrow [\ \text{V} \oplus \text{T} \oplus \text{si}\ ]$

However, as observed by Korostenskienė (2017), Šereikaitė (2017, 2022, 2023), and Bruening (2019), T is not the only portion of the word that *si* 'jumps over': it also does so with verbal categorizers (9), and nominal categorizers and associated ϕ-features (10).

(9) a. aug-in-a-**si**
grow-V-PRS.3-*si*
'grows (sth) for self'

b. *augsiina

(10) a. mat-ym-as-**is**[3]
see-n-SG.NOM-*si*
'the seeing of each other'

b. *matsiymas

To salvage the explanation, string-vacuous LD should be proposed for each of these cases. This account thus needs ad hoc machinery to explain how *si* is dislocated (Williams 2007). What emerges is that there is no morphological generalization to be made. *Si* is always peripheral when it is a suffix: morphological (or syntactic) approaches cannot capture that.

Other DM accounts differ from this proposal. Thus, while in Embick and Noyer's (2001) analysis *si* is initially linearized before all prefixes, Arregi and Nevins (2012) propose that in the initial linearization, *si* precedes the LP but not other prefixes, as in (11).

(11) Neg-Asp-***si***-LP-verb

The accounts that assume these initial linearization explain the final position of *si* after the LP, as in (*ne-*)*pa-si-mato*, as a Wackernagel (second position) effect: *si* jumps one adjacent constituent to the right. However, we will now show that there is in fact no way to characterize a rule for the position of *si* if it is assumed that it linearizes like in (11). Let us consider what the rule of the movement of *si* could be formulated as, assuming this initial linearization.

If it is proposed that *si* is an exponent that seeks to escape some left edge, then the following problem is encountered: *ne-* is enough to keep *si* in its place in some cases (*ne-si-mato*) but not others (*ne-pa-si-mato*). Hence, *si* behaves differently when the environment on its left is controlled for. It seems, then, that *si* might be sensitive to the LP, that is, the element to its right *si*. However, this does not explain why *si* moves rightwards when there is no LP, as in *mato-si*. Furthermore, both in *mato-si* and in *ne-si-mato*, the context to the right of *si* is V and T, but *si* is displaced in the former and not the latter. Hence, *si* behaves differently when the environment on its right is con-

trolled for. Thus, it is not possible to formulate a rule for the movement of *si* sensitive to the environment on its left or to its right, if the starting linearization is '*si*-LP-base'. The only possible generalization for this is the disjunction in (12).

(12) *Si* moves either when there is an LP, or when there are no prefixes.

This is unsatisfactory. Therefore, we conclude that any attempt to explain the distribution of *si* as a second-position-in-a-word effect on the *si*-LP-verb order fails. We are led to assume that in the initial linearization, *si* has to already be closer to the verbal root than any prefixes, like we propose in the next section.

To our knowledge, the only non-hierarchical accounts that describe the distribution of *si* correctly are Stump 2022, Šereikaitė 2017, and Kushnir 2025. Šereikaitė (2017) takes *si* to originate as the specifier of VoiceP. She builds on the DM literature on *si*, but proposes that it has two initial linear positions: "If the host has no prefix, -*si*- is linearized as a suffix after T. If the host has a prefix, -*si*- is linearized as a prefix" (Šereikaitė 2017:452). We find this undesirable: the linearization mechanism should trade the hierarchical relations for ones of linear precedence, and this proposal assumes a mechanism where the way Spec-VoiceP is initially linearized depends on the way other elements have already been linearized. In addition, Šereikaitė's account, as most other available accounts, makes no predictions with respect to the position of *si* in unprefixed derived forms without T, like unprefixed nominalizations, in which suffixal *si* is appended to the nominal inflection.

Stump (2022), working from a Word Formation Rules perspective, argues that *si* is always suffixal. When it occurs in the presence of a tensed verb, *si* suffixes to Tense inflection; when it occurs in the presence of an LP, *si* suffixes onto it. This captures the facts about the linear position of *si* with verbs, although, again, should need to add

extra machinery to capture the position of *si* is nominalizations without prefixes. In addition, this account misses some empirical facts about phonological interactions of *si* with its host. As we will argue in subsection 7.2, segmental effects show that *si* is not a suffix to a prefix when it occurs before the verb, *pace* Stump (2022).

2.2.2 Syntactic Accounts

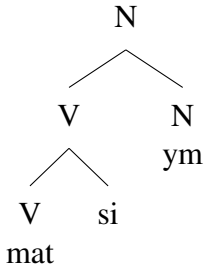
The syntactic accounts argue that the variable linear positioning of *si* is due to its variable positioning in the syntax.

Bruening (2019) proposes the following well-formedness condition:

(13) A Lithuanian V^0 that is [+Refl] must dominate a $Refl^0$ morpheme that is right-adjoined to the highest X^0 that is adjoined to V^0.

As such, *si* is adjoined to T, unless there is a higher head adjoined to V—such as an LP, in which case *si* adjoins to it. However, this account does not capture the facts from nominalizations (see (10)). Indeed, the nominalizing head is quite evidently not adjoined to V, but, if adjunction is at all involved here, the other way around, since nominalizers impose their category to the derived word. The structure of the nominalization of a *si*-reflexivized verb 'to see' is illustrated in (14). Bruening's condition forces *si* to be adjoined to V in this case, and therefore the ungrammatical **matsiymas*, reproduced below, is predicted. In addition to its descriptive inadequacy, Bruening's complex wellformedness condition is totally idiosyncratic to *si* and could arguably raise issues as far as language acquisition is concerned.

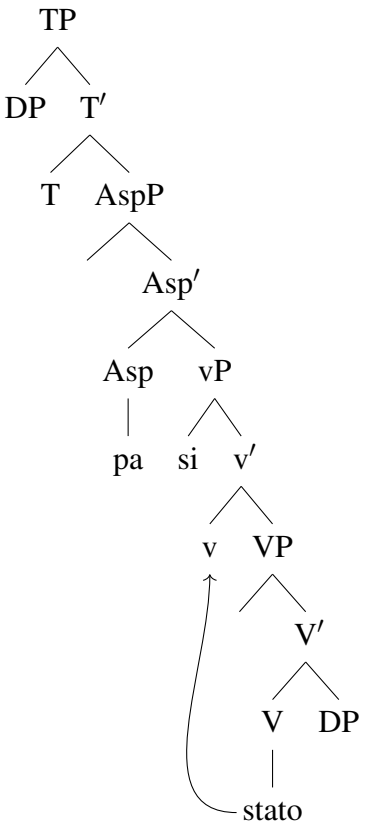
(14) a.

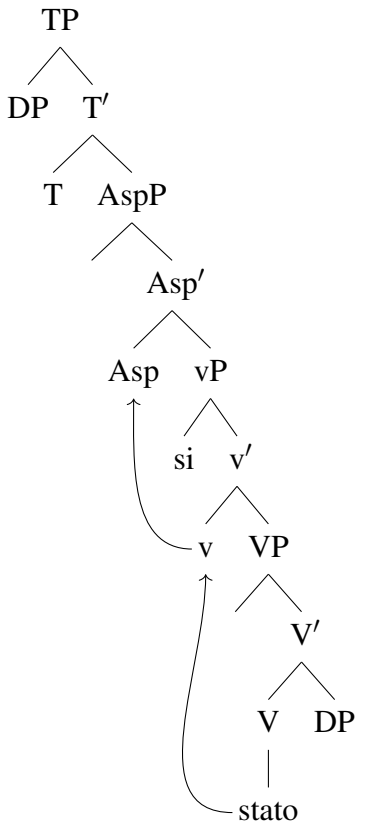


b. *mat-si-ym-as
see-*si*-NMLZ-SG.NOM

Korostenskienė (2014, 2016a, 2016b) explains the position of *si* by either allowing or blocking the movement of *si* and the verb up the tree. For her, *si* is introduced in the syntax, but in different positions, according to interpretation. For example, on her account of agentive *si*-intransitives, *si* originates in Spec-vP, while the head of Asp is filled by the LP. The verb moves up to v, if there is a LP in Asp, or to Asp, if there is no LP. This derives the correct linear order of *si*, in the syntax.

(15) a. *pa-si-stato* ‘builds for self’ b. *stato-si* ‘is building for self’





(from Korostenskienė (2016b))

Korostenskienė (2014, 2016a, 2016b, 2017) does not explicitly discuss how syntactic structures are linearized. However, two implicit points in her account are worth drawing attention to. First, the LP merges high, in Asp. Second, the verb moves up (to Asp or to v) to derive the correct linear order in the syntax. The evidence suggests, however, that the Lithuanian verb is linearized quite low. This is shown by Korostenskaja (2014), who applies Pollock’s (1989) tests such as manner, frequency, and epistemic adverb placement, yes/no questions, negation, and floating quantifiers. (16) demonstrates the verb’s relative order with respect to adverbs like ‘often’ and ‘on purpose’.

(16) Marija dažnai tyčia dainuoja, kad sunervintų kaimynus.
Marija often on_purpose sings that annoy neighbours
'Marija often sings on purpose, to annoy her neighbours.'

Example (16) shows that these adverbs precede the verb in Lithuanian. Taking the frequency adverb 'often' to be merged with AspP, and the agent-oriented adverb 'on purpose' to be merged below Asp, with a thematic VoiceP, we conclude that the verb does not linearize above Voice.[4] We thank an anonymous reviewer for bringing our attention to examples with compound tense, like (17). They provide further support for the claim that the lexical verb —namely, the active participle: *atsiskyręs* in (17)— is linearized below T, since T is already occupied by the auxiliary. Agent-oriented adverbs 'consciously' and 'on purpose' are between the auxiliary and the lexical verb, as expected for Voice-level adverbs.

(17) Jis yra sąmoningai / tyčia atsiskyręs nuo gyvenimo žemėje.
He is consciously / on_purpose separated from life on_Earth
'He has consciously/on purpose separated himself from life on Earth.'

If the inflected verb in Lithuanian does not move to Asp, Korostenskienė's account cannot be correct.

In addition, the evidence from the nature of the LPs suggests that they originate low in the structure, again contra Korostenskienė (2014, 2016b). These prefixes are of prepositional origin, and are attested in so-called weak satellite-framed languages (Balto-Slavic, Latin, Ancient Greek, Ancient Gothic, i.a.) (Acedo-Matellán 2016). Other languages—-strong satellite-framed, for example, Germanic—-have their free morpheme equivalents (e.g. English *in* in *step in*). Being located under Voice, LPs may have an effect on argument structure (introducing internal arguments) and contribute aspectual properties (telicity) as well as idiosyncratic meanings to the verb.

While some authors (e.g., Korostenskienė 2014, 2016b) have assumed the LP to originate in Asp, the standard way to treat these prefixes is to have them originate closer to the root (for Lithuanian see, e.g., Šereikaitė 2018). We develop further arguments and a concrete implementation in section 5.

Assuming *si* to be a pronominal clitic (but see section 3.2), a syntactic proposal for variable clitic placement can either involve the verb moving past the clitic, or the clitic moving past the verb. As we have shown, the former option will not work for Lithuanian, since the verb stays low and does not move past *si*. Alternatively, *si* could be moving past the verb if *si* is introduced lower in the structure, for instance, as the complement of V, as has been proposed for some instances of *si* (Korostenskienė 2016b). Its default position is then post-verbal. However, this syntactic position is incompatible with some properties of *si*. For instance, *si* appears in constructions with internal arguments like middles/unaccusatives (illustrated in (18)-(19)). If *si* originates as the complement of V, there would be then two internal arguments first merged in the same position—the derived subject and *si*.

(18) Šita duona lengvai pjauna-si.
this bread easily cut.PRS.3-*si*
'This bread cuts easily.'

(19) Mergaitė mato-si nuotraukoje.
girl.SG.NOM see.PRS.3-*si* picture.LOC.SG
'The girl is visible in the picture.'

It could be proposed that different occurrences of *si* are generated in different syntactic positions: some in Voice, some as the complement of V. The linear position of *si* would then have to be derived differently in each case. That is, even though the linear position of *si* is the same in the reflexive construction *prausiasi* 'washes oneself' and

in the middle construction *pjaunasi* 'cuts', it would be derived differently. In this case, the generalization is lost that, no matter the syntactic and semantic properties associated to *si*, or whether the verb is unergative or unaccusative, its linear position ignores them and only ever depends on the presence of prefixes. Finally, purely syntactic analyses like Korostenskienė's have no way of deriving the peripheral, rather than internal, position of *si* in nominalizations not involving prefixes. It emerges, then, that there is no purely syntactic generalization for the linear position of *si* either.

3 Empirical Generalizations

3.1 Linearization

The following two generalizations capture the linearization of *si*:

Generalization 1. In unprefixed words, *si* appears as the rightmost exponent, after any categorizers and inflectional morphology.

(20) a. mat-o-**si**
see-PRS.3-*si*

b. *__si__-mat-o
si-see-PRS.3

c. *mat-__si__-o
see-*si*-PRS.3

(21) a. aug-in-a-**si**
grow-V-PRS.3-*si*

b. *__si__-aug-in-a
si-grow-V-PRS.3

c. *aug-__si__-in-a
grow-*si*-V-PRS.3

d. *aug-in-__si__-a
grow-V-*si*-PRS.3

Generalization 2. In prefixed words, *si* is left-adjacent to the verbal root.

(22) a. pa-**si**-mat-o
LP-*si*-see-PRS.3

b. *__si__-pa-mat-o
si-LP-see-PRS.3

c. *pa-mat-__si__-o
LP-see-*si*-PRS.3

d. *pa-mat-o-**si**
LP-see-PRS.3-*si*

(23) a. ne-pa-**si**-mat-o
NEG-LP-*si*-see-PRS.3

b. *__si__-ne-pa-mat-o
si-NEG-LP-see-PRS.3

c. *ne-__si__-pa-mat-o
NEG-*si*-LP-see-PRS.3

d. *ne-pa-mat-__si__-o
NEG-LP-see-*si*-PRS.3

e. *ne-pa-mat-o-__si__
NEG-LP-see-PRS.3-*si*

(24) a. ne-be-pa-**si**-mat-o
NEG-PROG-LP-*si*-see-PRS.3

b. *__si__-ne-be-pa-mat-o
si-NEG-PROG-LP-see-PRS.3

c. *ne-__si__-be-pa-mat-o
NEG-*si*-PROG-LP-see-PRS.3

d. *ne-be-__si__-pa-mat-o
NEG-PROG-*si*-LP-see-PRS.3

e. *ne-be-pa-mat-__si__-o
NEG-PROG-LP-see-*si*-PRS.3

f. *ne-be-pa-mat-o-__si__
NEG-PROG-LP-see-PRS.3-*si*

An overarching generalization is that *si* is always adjacent to the inflected base.

Based on these facts, we propose that *si* is by default linearized as a prefix to the verbal base.

(25) (prefixes-)***si***-verbal base

Section 5 lays out the details of how we derive this linear order using Spanning.

It must be noted that two apparently prefixed verbs, *pasakoti* 'to narrate' and *pažinti* 'to know', differ from the scenario depicted thus far with respect to the linearization of *si*. Thus, in (26), *si* does not intervene between the 'prefix' and the root, as would be expected from the examples seen so far.

(26) a. pa(-)žin-ti-**s**
pa(-)know-INF-*si*
'to get acquainted'

b. pa(-)sak-o-ti-**s**
pa(-)say-V-INF-*si*
'to narrate (about oneself)'

These two verbs behave very differently from other LP-prefixed verbs in other re-

spects, and there is reason to suspect that *pa* in these verbs is not a prefix, but part of the root—but see Šereikaitė (2018, 2022) for an analysis of *pa-* in these cases as a nominal prefix, merged in an initial nominal cycle. Crucially, in Lithuanian LP-prefixed verbs cannot be prefixed with a further LP (see Kozhanov 2016, i.a.). However, the example in (27) shows that these verbs can (the dubious morphemic boundary is indicated with brackets).

(27) pa-pa(-)sak-o-ti
LP-*pa*(-)say-V-INF
'to narrate a little bit'

These two verbs also differ from LP-prefixed verbs in their aspectual properties. Verbs prefixed with LPs are normally telic (see Korostenskienė 2015:116, i.a.). However, *pa*(-)*sakoti* and *pa*(-)*žinti* are atelic, as illustrated for the former in (28).

(28) Pa(-)sak-o-jau pasaką {tris valandas / *per tris valandas}
pa(-)say-V-PST.1SG tale.ACC.SG {three.ACC hour.ACC.PL / *in three.ACC hour.ACC.PL}
'I told a tale {for three hours / *in three hours}.'

In light of this strong evidence, we follow Šereikaitė (2017) in assuming that in these verbs the sequence *pa* is not a prefix in the synchronic grammar, but rather part of the root. From this perspective, the behavior of *si* with these verbs turns out to be regular.[5]

To anticipate further analysis, we will propose that *si* stays in the position in which it is first linearized (as in (25)) if more prefixes are added, and it is displaced to the rightmost position of the prosodic word if not. This is in contrast to part of the DM literature, which assumes that *si* linearizes before the verbal base and some or all pre-

fixes, and the surface order of *si* is derived via Local Dislocation of *si* over a subword.

Another possible hypothesis is that *si* is by default linearized as a suffix, moving to a prefixal position whenever a prefix is added, as in (29).

(29) (prefixes-)inflected verbal base-***si***

This hypothesis has not been entertained by any author. It raises issues in cyclicity and lookahead. The suffixal position of *si* is also impossible to characterize in morphosyntactic terms, since *si* can appear after verbal inflection but also after nominal inflection, in nominalizations (see section 7.1).

3.2 The Syntax and Semantics of si

Despite some accounts that take *si* to be syntactically and semantically inert (Embick and Noyer 2001), there is evidence that *si* is related to argument structure and interpretation. As has been pointed out by, among others, Arregi and Nevins (2012:252–253), and Korostenskienė (2014:61), *si* has a range of uses that are typical of Indo-European reflexive morphology. Indeed, the addition of *si* is a productive way to form reflexives, reciprocals, anticausatives, and middles (see discussion in Geniušienė 1987). (30) illustrates this (examples from Utka et al. 2017 and Google Search).

(30) a. Ona parkuoj-a-si prie parduotuvės. *reflexive*
Ona park-PRS.3-*si* near shop
'Ona is parking (her car) next to the shop.'

b. Vaistas greitai absorbuoj-a-si į kraujotaką. *anticausative*
medicine quickly absorb-PRS.3-*si* to bloodstream
'The medicine is quickly absorbed into the bloodsream.'

c. O jums tai tikrai lengvai žygiuoj-a-si. *middle*
And you.DAT.PL then really easily hike-PRS.3-*si*
'And for you, hiking is really easy.'

d. Usnis su Dagiliu kažin ką šnibžd-a-si. *reciprocal*
U. with D. something whisper-PRS.3-*si*
'U. and D. were whispering something among themselves.'

The syntactic nature of *si* has not received much attention in the literature. Korostenskienė (2016b) assumes *si* to be an overt trace of the subject (as opposed to a valency-reducing element), and argues that *si* is a DP argument that originates either in Spec-vP (for those *si* that are coreferential with agentive subjects), Spec-VP (for experiencer subjects), or as a complement within VP (for Theme subjects). Arregi and Nevins (2012:253) assume *si* to be "the realization of a v category [...] that takes VP as its complement," which might be problematic, depending on what is meant by "v category," since *si* can cooccur with verbal categorizers like *-in* (see, e.g., (21)). Šereikaitė (2017), arguing from the formation of anticausatives with *si*, claims that *si* originates in the specifier of an expletive Voice head, while in later work (Šereikaitė 2023), she argues that it may be a head or a specifier. Kushnir (2025) assumes, without discussion, that *si* is the exponent of a Voice head itself.

We also think that *si* is best analyzed as the realization of a special Voice head, rather than as a pronominal specifier of Voice that ends up cliticized. Two of our arguments come from Wood (2015), who explores the properties of the Icelandic mor-

pheme -*st*, very close in semantics to Lithuanian *si* and also allowing reflexive, reciprocal, anticausative, and middle readings. Some of the arguments that Wood uses to claim that -*st* is a reduced clitic pronoun merged as the specifier of a Voice head (and other functional heads), rather than an inflectional head, can be used, reversely, to show that Lithuanian *si* represents an inflectional head (Voice) rather than a pronoun. First, Wood (2015:69–70) argues that -*st* must have some ϕ-feature, since, when it is present on a finite verb, no person distinctions on the verb can be made in the singular, suggesting that "-*st* can intervene for person agreement". This is unlike Lithuanian *si*, which can be added as such to any person-inflected form of the verb.

(31) prausiuo-**si**, prausie-**si**, prausiamė-**s**, prausiatė-**s**
wash.PRS.1SG-*si* wash.PRS.2SG-*si* wash.PRS.1PL-*si* wash.PRS.2PL-*si*

If *si* does not intervene for person agreement, its status as a pronoun is undermined. An anonymous reviewer asks whether these facts could be compatible with a type of analysis where *si* is an object pronoun lacking ϕ-features and originating lower in the structure than Voice, where the subject with ϕ-features is located. Whereas this alternative scenario would indeed account for the featural inertness of *si* shown in the above example, there is evidence, kindly pointed out by the same reviewer, that seems incompatible with the analysis of *si* as originating in an argumental position. Such evidence involves the availability of proxy readings for the reflexive marker, that is, readings where the entity denoted by the reflexive is not exactly the same as that denoted by the subject, but a proxy or copy thereof. Korostenskienė (2017:468) provides the following contrast between a *si*-prefixed verb, on the one hand, and its unprefixed counterpart combined with the free anaphor *save*, on the other:

(32) a. Saulius už-si-dengė.
Saulius.NOM LP-*si*-cover.PST.3
'Saulius covered himself.'

b. Saulius už-dengė save nuotraukoje.
Saulius.NOM LP-cover.PST.3 self photo.LOC.SG
'Saulius covered himself (= his image) on the photo.'

Only the free anaphor licenses a proxy reading denoting not Saulius himself, but his image on the photograph. Marelj and Reuland (2013) convincingly argue, on the basis of Slavic reflexives, that proxy readings are only possible with reflexive elements merged in argumental positions. If that is the case, *si* cannot be analyzed as an object pronoun. Its analysis as a Voice head, however, is compatible with the evidence just shown.

Second, Wood (2015) makes a crosslinguistic observation that "nominalizations can often include derivational affixes, but not argument clitics" (Wood 2015:80). As it turns out, Lithuanian *si* is perfectly compatible with nominalizations, suggesting that it is not a pronominal clitic. This will be shown more extensively in section 7.1, where we discuss nominalizations with *si* and provide an analysis. What we want to emphasize here is that, were Lithuanian *si* an argument clitic, its compatibility with eventive nominalizations would not be aligned with what we find in other languages with well-established argument clitics, like Romance (Kayne 1975:186, Pesetsky 1995:99, apud Wood 2015:80), and Slavic (Franks and Tracy 2000:273ff). We provide a Portuguese example (our own) involving a reflexive clitic and a Serbo-Croatian example (from Ž. Bošković, apud Franks and Tracy 2000:273) involving a non-reflexive object clitic.[6]

(33) *a destruição-**se** em cinco dias
the destruction-REFL.3 in five days
Int.: 'the destruction of itself in five days'

(34) *prestavljanje **ga** Marija
introduction 3SG.M.ACC Marija.DAT
Int.: ‘the introduction of him to Marija’

Our last argument for the claim that *si* is the exponent of a Voice head, rather than a pronominal clitic, is phonological, and anticipates one of our predictions explored in section 7.2. In a nutshell, suffixal *si* behaves unlike an enclitic in that it is not associated with a word-boundary to its left. For instance, the monophthongization of the diphthong /iɛ/ happens word-finally (35a), even before the encliticized particle *gi* ‘after all, indeed’ (see Ruskan 2019), as shown in (35b). But it is not observed before suffixal *si*, where the diphthong is preserved (35c).

(35) a. Kel**i** ranką.
raise.PRS.2SG hand.ACC
‘You raise your hand.’

b. Kel**i** gi anksti šiandien!
raise.PRS.2SG *gi* early today
‘You’re getting up early today, huh!’

c. Kel**ie**-si anksti.
raise.PRS.2SG-*si* early
‘You get up early.’

This evidence suggests that, rather than a clitic appended onto the verb, *si* is one more of the exponents making up the verbal word, like the future suffix -*s*, which also preserves stem-final diphthongs (see footnote 21).[7]

But what kind of Voice head(s) does *si* correspond to? We adopt Labelle’s (2008) proposal for French *se*. *Se* may realize a Voice head that introduces an agent variable and identifies it with an unsaturated variable contributed by the phrase with which it merges.

(36) $[\![\text{Voice}_{\text{refl}}]\!] = \lambda P \lambda x \lambda e.P(e,x)$ & Agent(e,x)

Simplifying somewhat, when this Voice combines, via Functional Application, with something like λyλe.wash(e) & Theme(y,e), that is, a relation between entities and events of washing such that the entity is the Theme, the result is a relation of individuals and events of washing such that the individual is both the Theme and the Agent.

Reflexives and reciprocals may be transitive.

(37) Ona pa-si-statė namą kaime.
Ona LP-*si*-build.PST.3 house.ACC countryside.LOC
'Ona built herself a house in the countryside.'

(38) Tie du studentai ap-si-keitė knygomis.
Those two students LP-*si*-change.PST.3 book.INSTR.PL
'Those two students exchanged books.'

For these cases, we assume, always following ideas in Labelle 2008, that a null Applicative head merged above vP (Pylkkänen 2008) introduces the unsaturated variable, a Benefactive, that $\text{Voice}_{\text{refl}}$ ends up identifying with the Agent.

Finally, French *se* and Lithuanian *si* in anticausatives and middles involve a Voice head, here labelled $\text{Voice}_{\text{antic}}$, that also combines with vPs with an unsaturated variable, but that does not add an Agent θ-role (see also Schäfer 2008 and Wood 2015, i.a., for the idea that Voice can be expletive).

(39) $[\![\text{Voice}_{\text{antic}}]\!] = \lambda P \lambda x \lambda e.P(x,e)$

Again simplifying a bit, when this Voice head is combined with something like λyλe.absorb(e) & Theme(y,e), the result is semantically identical to the latter. As a result, the DP merging as specifier of $\text{Voice}_{\text{antic}}$ will be understood as the Theme (see (30b)).

All in all, we assume that *si* is the realization of a Voice head whose sister con-

tributes an unsaturated variable over individuals. For the time being, and with the hope of accomplishing a unified analysis in the future, we surmise that such characterization corresponds to either $\text{Voice}_{\text{refl}}$ or $\text{Voice}_{\text{antic}}$.[8]

4 Theoretical Assumptions

We assume a variant of the Y-model of the architecture of grammar, following many minimalist approaches. We adhere to a nonlexicalist stance, where words do not come into the syntactic component already assembled by another component, but their parts are assembled in the syntax. We assume cyclic spell-out, where the merger of certain heads in the syntax triggers transfer to PF, meaning that the derivation up to that point is spelled out and treated as a phonologically relevant unit, as in phase theory (Chomsky 2000, 2001).

This section presents our theoretical assumptions regarding spell-out, linearization, and the syntax-prosody interface.

4.1 Spanning and Linearization

The analysis presented here uses Spanning as the theory of assignment of exponents to syntactic representations (Williams 2002, Bye and Svenonius 2012, Svenonius 2016, Acedo-Matellán and Kwapiszewski 2026). Exponents are assigned to *spans*, which are sequences of heads in a head-complement relation. For example, the derivation $[_{\text{TP}}$ T $[_{\text{VoiceP}}$ Voice $[_{\text{vP}}$ v]]] involves the spans ⟨T, Voice, v⟩, ⟨T, Voice⟩, ⟨Voice, v⟩, as well as the singleton spans ⟨T⟩, ⟨Voice⟩, ⟨v⟩. Each of these spans can be spelled out with a single exponent.

The definition of span predicts that specifiers and adjuncts form spans of their

own, and are not part of the span of which they are dependents. For example, a DP sitting in the specifier of TP and containing the heads D, Num, and n, forms a span, ⟨D, Num, n⟩, that is not part of the span that contains T.

The lexicon is accessed cyclically for the retrieval of exponents. In particular, lexical access is regulated by a diacritic w with which certain heads are marked (Svenonius 2016). A span whose hierarchically highest head is marked with the w diacritic is a w-span. Each w-span determines a cycle of lexical access, where a single exponent or several exponents may be retrieved to spell it out. Each such cycle generates a minimal word, free or not, where idiosyncratic relations of form and meaning may be attested. (40) would be an example of such a w-span.

(40) ⟨T^{w}, Voice, v⟩

Standard phasal heads like C seem to determine spellout domains universally: the head of their complement is endowed with w in all languages. But spellout domains can be smaller than this, the size varying crosslinguistically. For example, Svenonius (2016) shows examples of languages where the plural marker corresponds to a spellout domain separate from that of the noun, a different word, unlike in English.

A standard assumption is that the assignment of exponents and their linearization proceeds from the bottom up, starting from the most embedded span. Following Mirror Theory (Brody 2000), within a w-span, exponents are linearized as suffixes unless they contain an instruction to trump this default mechanism (Bye and Svenonius 2012). Thus, within the minimal word, the hierarchical relation of embedding is by default traded for a relation of linear precedence: if X dominates Y and each is spelled out by a different exponent, x and y, then x follows y when linearized (Svenonius 2016), as (41) illustrates.

(41) a. ⟨X, Y⟩

b. [... y-x ...]

The sentential position of the phonological units generated from spans is regulated by a different diacritic: @ (Brody 2000). A span, comprised of one or more w-spans, endowed with the @ diacritic on head X is called an @-span. Each @-span generates a word positioned after all other words generated from @-spans higher than X, including all specifiers and adjuncts that are merged above X. For example, the verbal span in English has the @ diacritic on Voice, which means that the verb will be linearized after any other spans (including all specifiers and adjuncts) that are merged above Voice. Thus we see frequency adjuncts like *often* (adjoined to AspP, which is above Voice) preceding the verb in English. In French, however, @ is on T, which means that the ⟨T, Voice, v⟩ span will be spelled out and placed before the frequency adjunct (*souvent* 'often'), which forms a span of its own and is merged below T (Pollock 1989).

(42) a. ⟨T^{w}, $Voice^{@}$, V⟩: John often **kisses** Mary.

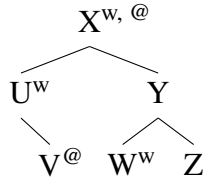
b. ⟨$T^{w,\ @}$, Voice, V⟩: Jean **embrasse** souvent Marie.

It may be the case that an @-span has specifiers or adjuncts that are not @-spans. Such spans cannot, of themselves, determine a position within the sentence and their position is dependent on the word generated from the @-span of which they are dependents (Svenonius 2016, Acedo-Matellán and Kwapiszewski 2026). The addition of dependents onto the linearized @-span also takes place bottom-up, the lowest @-defective dependent span being added first.

To sum up, the linearization algorithm determines the boundaries of linearized

@-spans, represented below with square brackets; the linear precedence relations between @-spans—either alone or hosting any positionally dependent w-span—represented below with the ⌢ symbol; the boundaries and precedence relations between remaining w-spans, represented below via the hyphen; and the linear precedence relations between exponents, represented here by the linear order of exponents. Finally, it preserves prosody-relevant diacritics, as we will see in the case of Lithuanian *si*. To illustrate, and assuming that the trivial spans ⟨U⟩, ⟨V⟩, ⟨W⟩, ⟨X⟩, ⟨Y⟩, and ⟨Z⟩, correspond to the single exponents *u*, *v*, *w*, *x*, *y*, and *z*, respectively, the syntactic representation in (43a) is mapped onto the linearized representation of (43b).

(43) a.



b. [*vu*]⌢*w*-[*zyx*]

The representation in (43b) contains two independent units, [*vu*], and *w*-[*zyx*], the former preceding the latter. Brackets represent the boundaries of linearized @-spans: [*vu*], and [*zyx*]. Other linearized w-spans are positionally dependent on @-spans: thus, *w* is positionally dependent on [*zyx*]. This representation is ready to be interpreted by the prosodic component.

4.2 Prosodic Structure and the Syntax-Prosody Interface

After the syntactic derivation has been fully linearized, it enters the prosodic component, which builds a fully fledged prosodic representation out of it. We assume the theory of prosodic constituency in Ito and Mester (2009). The particular constituents that we will refer to are the *prosodic word* (ω) (also called *phonological word*), the *phonological phrase* (ϕ) and the *intonational phrase* (ι).

(44) [ι [ϕ [(ω) [ω]]]]

As will become apparent in section 6, certain constraints regulate the alignment of the linearized representations delivered by syntax and the prosodic constituents. For instance, an optimal prosodification is that where the left and right boundaries of a linearized @-span (indicated with brackets on the linearized representation) are aligned, respectively, with the left and right boundaries of a prosodic word. Different circumstances may prevent this optimal alignment to be effected. For instance, @-spans not containing a root, like the English future auxiliary *will*, may destress and encliticize.

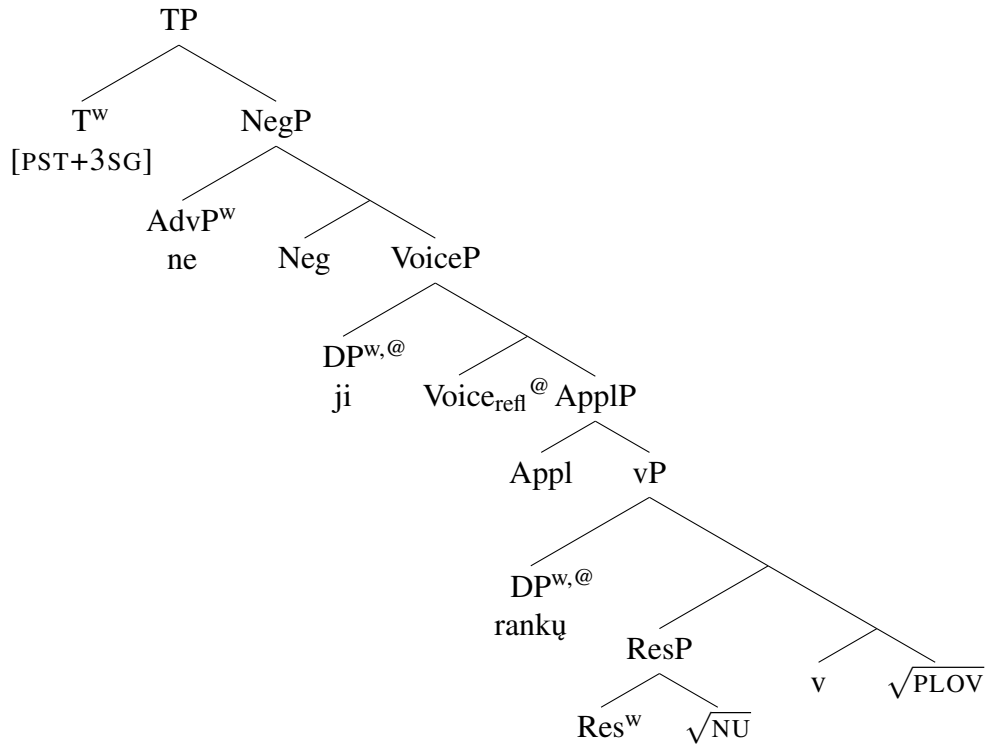
While linearized @-spans are mapped, by default, onto (minimal) prosodic words, elements attached to them by the linearization algorithm (mapped from @-defective specifiers and adjuncts), are parsed as proclitics, in Lithuanian, to this prosodic word, iterating the prosodic word category (Ito and Mester 2009).

5 The Default Prefixal Position of *si*

Let us illustrate how the default linearization of *si* as proposed in (25) is derived, using the example in (45).

(45) Ji ne-nu-si-plov-ė rankų.
she.NOM NEG-LP-*si*-wash-PST.3 hand.GEN.PL
'She did not wash her hands.'

(46)



The verbal root $\sqrt{\text{PLOV}}$ is first merged with v, introducing, in the semantics, a predicate of eventualities of washing. The LP *nu-* corresponds to ResP, comprising the stative head Res (Ramchand 2008, Folli and Harley 2020) and the root $\sqrt{\text{NU}}$, and denoting a predicate of states (of "completion"). We follow a well-established tradition in the analysis of resultative (lexical) prefixes in Slavic languages (Svenonius 2008, Tolskaya 2018, Kwapiszewski 2022, Acedo-Matellán and Kwapiszewski 2026), Latin and Ancient Greek (Acedo-Matellán 2016, Asyllogistou 2019, Acedo-Matellán and Kwapiszewski 2026), and resultative prefixes and particles in Germanic languages (Hoekstra and Mulder 1990, Mateu 2008, Ramchand 2008, Wood 2015, Folli and Harley 2020, Tan 2024), whereby these elements are integrated in phrasal constituents denoting a location or state resulting from the event (for the particular idea that they involve a root, see Acedo-Matellán 2010, Haselbach and Pitteroff 2015, Wood and

Marantz 2017 and Tan 2024, i.a.). One of the standard tests for the predicative nature of LPs is the ability to introduce unselected objects. Lithuanian LPs are no exception, as the following prefixed verb based on *nu-* and the intransitive verb *miegoti* 'sleep' illustrates:[9]

(47) Ji *(nu-)miegojo šoną.
she LP-sleep.PST.3 side.ACC
'She slept on her side so that it became numb.'

ResP has been independently assembled and is then merged with v. Since this is a second application of Merge, ResP is, necessarily, a specifier (Acedo-Matellán and Kwapiszewski 2026; see also Starke 2018). The denotation of the vP is as follows:

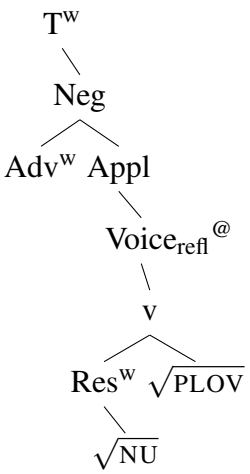
(48) $[\![\text{vP}]\!] = \lambda x \lambda e \exists s.\text{wash}(e) \,\&\, \text{Holder}(x,s) \,\&\, \text{cause}(e,s) \,\&\, \text{completion}(s)$
"a relation between entities (x) and washing events (e) leading to a state of completion (s) such that x is the holder of s"

The object *rankų* 'hands' is merged as a second specifier of vP and saturates the variable corresponding to the holder of the state resulting from the washing event. Above vP, a null Appl head is merged, introducing a variable over benefactives or entities affected by the eventuality ('she washed her own hands/for herself'). $\text{Voice}_{\text{refl}}$ is then merged, introducing the external argument *ji* 'she' and identifying it, as the Agent, with the unsaturated variable introduced by Appl, yielding the reflexive interpretation (see section 3.2). That this is the right analysis of *si*, that is, as a Voice head, rather than as a pronominal argument, is supported by the fact that (45) does not license a proxy reading involving the washing of the hands of a statue, say, rather than her own hands. We abstract away from further movement of the external argument in this representation. For the sake of simplicity, we are also not representing certain functional

nodes like Asp. Above Voice, we find the functional projection Neg, with an adverbial specifier that spells out as the prefix *ne-*. This projection triggers the genitive marking on the object *rankų* (Sigurðsson and Šereikaitė 2024). Above Neg, T is merged.

Disregarding the nominal projections corresponding to the subject and the object, the spans licensed by this representation are shown in (49).

(49)



One span corresponds to the verbal extended projection $\langle T^{w}$, Neg, Appl, $Voice_{refl}{}^{@}$, v, $\sqrt{PLOV}\rangle$. Another span is $\langle Res^{w}, \sqrt{NU}\rangle$. The third span is $\langle Adv^{w}\rangle$.

Let us see how linearization proceeds, starting out with the first, main, span, corresponding to the inflected verb. This is an @-span. First of all, note that, since @ is at Voice (see footnote 4), the verbal span will be linearized after the subject (merged above Voice) but before the object (merged below Voice). The verb itself receives exponence starting from the bottom-up. The algorithm first spells out the subspan $\langle$v, $\sqrt{PLOV}\rangle$ as /plov/.

(50) plov

The next span to receive an exponent is $\langle Voice_{refl}\rangle$. This exponent is *si*, which exhibits special linearization properties. For the time being, we concentrate on the fact that

the linearization algorithm concatenates *si* before rather than after the previously linearized output or base. The trumping of the default suffixation procedure is encoded in a diacritic readable to the syntactic linearization mechanism, which we represent simply with a suffixed hyphen.

(51) $\langle \text{Voice}_{\text{refl}} \rangle \Leftrightarrow$ *si-*

After *si* is linearized, we obtain the following sequence:

(52) ***si***-plov

The two subspans ⟨Appl⟩ and ⟨Neg⟩ receive zero exponents.

(53) ***si***-plov-∅-∅

The remaining span ⟨T⟩ is exponed as /e/ and suffixed, by default. The w-span has been fully linearized. This is an @-span, as per the @ diacritic on one of its subspans; its boundaries are represented with brackets.

(54) [*si*-plov-∅-∅-**e**]

This linearized @-span will be optimally mapped as a prosodic word in the prosody. What about the LP, that is, ResP, which is a specifier to v? It does not form a span with the verb, but Res is endowed with the w diacritic in Lithuanian, as in Ancient Greek or Slavic (Acedo-Matellán and Kwapiszewski 2026). It therefore triggers an independent cycle of lexical access. The exponent *nu* is matched to the whole span $\langle \text{Res}^{\text{w}}, \sqrt{\text{NU}} \rangle$. Since this span is @-defective, it cannot be linearized as a free-standing word and it is added before the base, which the prosodic component will optimally interpret as procliticization.

(55) **nu**-[*si*-plov-∅-∅-e]

In this analysis, therefore, lexical and other prefixes are attached to the fully linearized verb, including *si*. This accounts for one of the generalizations on the linearization of *si*, namely, that, as a prefix, it is immediately adjacent to the verbal base, appearing after any other prefix. Note that the only two assumptions particular to *si* that we have made are 1) that *si* is the exponent for $\text{Voice}_{\text{refl}}$, as the syntax-semantics suggests anyway; and 2) that it is marked as a prefix, trumping the default suffixation algorithm. However, all theories must stipulate some dimension of the linearization of *si*.

The linearized sequence nu-[*si*-plov-∅-∅-e] already exhibits the same linear order as the surface representation, unlike in other proposals where the initial linearization is *si-nu-plove*, and *si* has to undergo other post-syntactic movement processes to derive the surface linear order (Embick and Noyer 2001, Arregi and Nevins 2012).

Since Voice is part of the same @-span as the verbal base, the prediction of our algorithm is that it will linearize closest to the verb, after all prefixes. This includes, of course, those prefixes that are linearized before LPs like *nu*-, that is, so-called superlexical prefixes: *te*- (restrictive, among other interpretations), *ne*- (negation), *be*- (continuative, among other interpretations)—see Arkadiev (2012) and Korostenskienė (2022). Based on their meaning and motivated by crosslinguistic evidence (Mayali (Gunwinjguan), as described by Evans 1995), we assume that these prefixes originate above Voice (see also Korostenskienė 2017 and Šereikaitė 2017), and at least *te*- and *be*- are A-quantifiers, and hence adjuncts.[10] Thus, in (45) we assume that *ne*- corresponds to an AdvP, merged as the specifier of Neg. Since specifiers and adjuncts are not part of the main @-span, they are spelled out independently. We assume that these adverbs bear the w diacritic, which effectively allows them to initiate a lexical

cycle of their own, but are @-defective spans, and hence must be appended before the sequence generated by the main @-span and any other dependent merged lower than them. In the case at hand, the span $\langle \text{Adv}^{\text{w}} \rangle$ undergoes spellout as *ne* and is linearized as preceding the prefix *nu-*, since it is a dependent of the same @-span and it is merged higher than the span that generates *nu-*.

(56) **ne**-nu-[*si*-plov-∅-∅-e]

Regarding derivations where *si* surfaces as a suffix (*plove-si*), the initial linearization for these is as in (55), but without the prefix.

(57) [*si*-plov-∅-∅-e]

This is the sequence derived via the linearization algorithm in combination with the prefixal diacritic on *si*. It is delivered as such to the prosodic stage of phonological computation, where the sequence *plovesi* finally obtains. This is explored in detail in the next section.

6 *Si* as a Suffix: The Prosodic Explanation

Second-position effects are observed in many languages, including Ancient Greek (Taylor 1996), Serbo-Croatian (Radanović-Kocić 1988, 1996), Romanian, Bulgarian, Macedonian, Slovenian, Polish (Bošković 2001), Hittite (Garrett 1996), Chamorro (Chung 2003) and Irish (Bennett et al. 2016), to name a few (for a collection of works on second-position elements, see Halpern and Zwicky 1996). A prosodic explanation has been applied to many of them, including sentential second-position effects (Halpern 1992, i.a.). For the sake of comparison, and based on our acquaintance with the language, we concentrate on European Portuguese (EP). We observe a parallelism

between the linearization of Lithuanian *si* and that of pronominal clitics in EP. Building on this comparison, we propose that *si* is sensitive to the left edge of the maximal projection of a prosodic word.

6.1 European Portuguese Pronominal Clitics

In EP, pronominal clitics also appear either postverbally or preverbally, depending on surrounding material. These pronouns must appear as enclitics to the verb in verb-initial sentences, with preverbal referential subjects and other types of material that are arguably topical, like dislocates and frame-setting adverbials. Examples (58)-(61) illustrate these environments.

(58) Viu-**o** a Maria.
saw-him the Maria
'It was Maria who saw him.'

(59) A Maria viu-**o**.
the Maria saw-him
'Maria saw him.'

(60) As crianças viu-**as** ontem.
the children saw-them yesterday
'She saw the children yesterday.'

(61) Hoje viu-**o**.
today saw-him
'Today she saw him.'

Clitic pronouns must appear as proclitic when preceded by non-referential subjects, aspectual adverbs, sentential negation, wh-elements, and complementizers. These are illustrated in (62)-(66).

(62) Ninguém/Alguém **o** viu.
nobody/somebody him saw
'Nobody/Somebody saw him.'

(63) A Maria já **o** viu.
the Maria already him saw
'Maria already saw him.'

(64) A Maria não **o** viu.
the Maria not him saw
'Maria didn't see him.'

(65) Quem **o** viu?
who him saw
'Who saw him?'

(66) A Maria diz que **o** viu.
the Maria says that him saw
'Maria says that she saw him.'

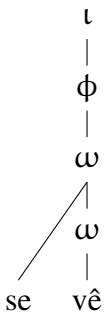
As observed by Barbosa (1996), there is nothing syntactic or morphological unifying the 'proclisis triggers'. This author proposes that in the syntax, the pronominal clitic moves to a position adjoined to TP. [11]

In Barbosa's account, the syntactic constituent TP corresponds, prosodically, to the intonational phrase, ι, thus excluding subjects (59), dislocates (60), or frame-setting adverbials (61), which map onto ιs of their own. If there is no other material before the clitic within the same ι, the clitic then finds itself at the left edge of ι, which, as Barbosa (1996) proposes, is forbidden:

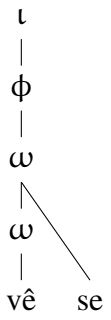
(67) *[$_{\iota}$ cl X]

When the clitic would occur in such an environment, as shown in (68a), it has to *tuck in* at the right edge of the prosodic word that is right-adjacent to it, namely, the verb, as in (68b).[12] If the derivation contains something else at the left edge of ι, like the low adverb *já* 'already', the clitic is protected from the left edge of the intonational phrase, as in (69), and no reordering takes place.

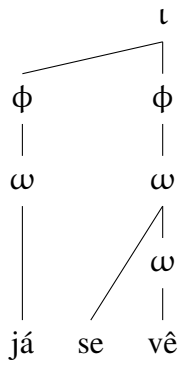
(68) a. ***se** vê
REFL.3 see.PRS.3SG
Int.: 'is visible'



b. vê-**se**
see.PRS.3SG=REFL.3
'is visible'

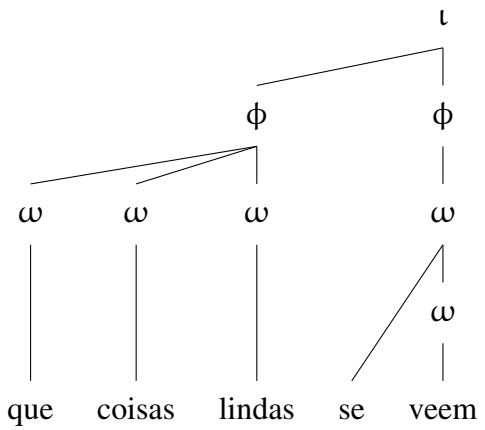


(69) já **se** vê
already REFL.3 see.PRS.3SG
'is already visible'



That the intonational phrase, and not a smaller constituent (e.g. ϕ), is the relevant constituent can be clearly seen in examples like (70). Here, the wh-phrase *que coisas lindas* 'what beautiful things' cannot plausibly be contained in the same phonological phrase as *se veem*, and yet, *se* is protected from the left edge. If a phonological phrase can protect *se* from the left edge, we can conclude that it is the left edge of a larger constituent, that is, the ι, that *se* cannot be aligned with.

(70) Que coisas lindas **se** veem?
what things beautiful REFL.3 see.PRS.3PL
'What beautiful things are visible?'



6.2 Lithuanian

We propose that Lithuanian *si* is affected by a constraint akin to that involved in EP pronominal clitics, only referring to a different prosodic constituent: *si* cannot occur at the left edge of a maximal prosodic word (see Šereikaitė 2022 and Kushnir 2025 for similar observations). To prevent that situation, *si* ends up at the right boundary of the prosodic word, deriving the suffixal position attested when no other prefix cooccurs.

The prosodic constituent with whose left edge Lithuanian *si* cannot be aligned must evidently be smaller than in Portuguese, since not being aligned with the left edge of an ι does not necessarily salvage *si*. For example, free morphemes such as adverbs like *jau* 'already' cannot protect *si* from the left-edge effect in Lithuanian, as they can with EP clitics.

(71) a. *jau **si**-mat-o
already *si*-see-PRS.3
Int.: 'is already visible'

b. jau mat-o-**si**
already see-PRS.3-*si*
'is already visible'

The question now is whether *si* is avoiding the left margin of the phonological phrase or that of an even lower-level constituent, that is, the prosodic word. The ev-

idence from sentences including verbal auxiliaries points to the latter. When an unprefixed verb with *si* appears after a stress-bearing auxiliary like *esu* 'be', *si* cannot appear word-initially, as shown in (72). Under the natural assumption that the stressed auxiliary and the lexical verb form a prosodic word each, both would be contained within a phonological phrase. Yet, the auxiliary does not allow *si* to remain word-initial. Therefore, the relevant constituent for the Lithuanian constraint must be smaller than ϕ, that is, ω.

(72) a. Aš esu mok-ęs-**is** lotynų.
I be.PRS.1SG learn-PTCP.PST.NOM.M.SG-*si* Latin.ACC
'I have studied Latin.'

b. [$_{\phi}$ [$_{\omega}$ esu] [$_{\omega}$ mokęs-is]]

c. *esu simokęs

As a matter of fact, derivations with prefixes reveal that we need to be more precise: the relevant constituent should be ω^{max}, and not ω. Since the prefixes are prosodically dependent on the verb (e.g., they cannot bear an independent stress), they form a single prosodic word with the verb. Following Ito and Mester (2009), who show that languages can refer to maximal prosodic words in their phonologies, we assume that prefixes recursively extend the projection of ω.

(73) [$_{\omega^{max}}$ prefix [$_{\omega}$ *si*-verb]]

This prosodification is predicted by the algorithm described in subsection 4.1: the verbal base up to Voice (i.e., *si*) corresponds to an @-span, which maps, by default, onto a prosodic word. Any @-defective dependents, that is, the prefixes, are appended as proclitics, forming successive segments of ω. In any case, when *si* occurs with prefixed verbs, it is evidently still aligned with the left edge of an ω, as seen in (73). Yet,

it remains preverbal. This illustrates that the relevant constraint for *si* refers to not just ω, but the maximal ω: *si* must not appear at the left edge of an ω^{max}.

(74) *[$_{\omega^{max}}$ *si* X]

But how is this prosodic alignment constraint encoded in the grammar, and, concomitantly, how is the suffixal order derived? Before the advent of Optimality Theory (OT), Halpern (1992:81) proposed a Prosodic Inversion rule to account for second position effects at the sentential level in different languages, including Serbo-Croatian and Ancient Greek. Other authors, like Radanović-Kocić (1996) for Serbo-Croatian clitics, Taylor (1996) for Koiné Greek clitics and Hale (1996) for the conjunction *u* 'and' in Vedic, have also proposed a movement rule (a prosodic "flip"). Applied to Lithuanian *si*, such a rule could look like the following:

(75) [$_{\omega^{max}}$ *si* X] → [$_{\omega^{max}}$ X *si*]

(75) would be a readjustment rule affecting *si* at the prosodic level. It would allow us to dispense with any diacritic marking this lexical item beyond the one enforcing prefixation at the syntactic linearization stage (see section 5). However, as a rule, it has no explanatory power. In particular, it stipulates that *si*, in order to escape the left edge of an ω^{max}, encliticizes to an adjacent ω. But the category ω seems to be quite unmarked as a clitic-host, crosslinguistically (Booij 1983, Zwicky 1985, Nespor and Vogel [1982] 2012), and this fact should therefore not be captured by a language-specific rule. In other words, and to illustrate the point with an example, there is nothing in a rule-based system precluding the formulation of a rule that, say, inserts quirky exponents like *si* five syllables to the right of the left boundary of the prosodic word. But languages do not work that way.

An OT-style account does away with these shortcomings, as it relies on a Lithuanian-specific ranking of universal constraints interpreting the representation delivered after linearization.[13] The analysis we develop here is based on that put forward by Bye and Svenonius (2012) for what they call *antitropal exponents*, that is, exponents that appear in positions not equivalent to those determined by the syntactic linearization algorithm.[14] In addition to infixes, Bye and Svenonius's (2012) account covers clitics similar to Lithuanian *si*, like the Latin conjunction *que* 'and', which must encliticize onto a right-adjacent ω. Antitropal exponents are marked with a diacritic indicating the type of prosodic edge with which the exponent cannot be aligned. In the case at hand, and using Bye and Svenonius's formalism, this diacritic, readable only to the phonology, identifies *si* as shunning the left edge of a maximal prosodic word (ω^{max}).

(76) $[_{\omega^{max}} \nsim$

To ensure that antitropal diacritics are respected, Bye and Svenonius (2012) formulate a high-ranking, anti-alignment constraint, IDENT[antitropal], which penalizes candidates in which the antitropal exponent appears in the position forbidden by the diacritic.[15] In addition to IDENT[antitropal], two other constraints are relevant in determining the position of the antitropal exponent. O-CONTIGUITY ('No intrusion') militates against infixation and assigns a violation for every output string exhibiting a non-edge-aligned internal segment that does not correspond to any segment of the corresponding input string. In turn, the constraint LINEARITY ('No metathesis') checks the effects of the displacement of material with respect to the input, assigning a violation for every instance of a linear precedence relation between two segments of the input that is not preserved in the output.

In addition to these constraints, we add two other constraints that ensure the right

mapping between syntactic and prosodic constituents. One such standard constraint is that establishing that the edges of lexical words must match those of prosodic words: see, for instance, Ito and Mester's (2009:142) LEX-TO-ω(L/R). However, in a neo-constructionist framework such as the one we adopt, the concept of lexical word has no meaning. In addition, we assume a model in which representations are linearized and provided with relevant boundaries by the syntax, before the prosodic component. Thus, we propose to reformulate the relevant constraint as one showing sensitivity to the edges of linearized @-spans: @-TO-ω(L) and @-TO-ω(R). They assign a violation for every instance of a linearized @-span whose left or right edge is not aligned with a left or right edge of an ω. In what follows, we will make use of @-TO-ω(L) but not @-TO-ω(R), as its presence is not relevant in the evaluation of candidates.
In turn, the proliferation of ω-boundaries facilitated by the alignment constraints is checked by NO RECURSION (Ito and Mester 2009:145), which penalizes each additional parse of any element into the same prosodic category. We propose the following ranking of the five constraints discussed:

(77) IDENT[antitropal], O-CONTIGUITY, @-TO-ω(L) ≫ LINEARITY, NO RECURSION

According to (77), the first three constraints are unranked relative to one another. Likewise, the last two constraints are unranked relative to one another. However, the first three constraints rank higher than the last two. This rather coarse-grained ranking is, as we will show in the tableaux below, adequate to generate the correct prosodic outputs for unprefixed and prefixed forms including *si*. We remain agnostic as to whether extending the analysis to other domains of Lithuanian phonology would require additional ranking relations among these constraints.

Let us consider first a base with no prefixes: *matosi* ‘is/are visible’. The input to prosody, that is, the output of the linearization algorithm, is [*si*-mat-o], where square brackets identify the edges of a linearized @-span and italicized *si* must be understood as representing [$_{\omega^{max}}$ ≁ *si*, that is, the exponent together with its antitropality diacritic.

(78) *Derivation of unprefixed forms*

[*si*-mat-o]	ID[ant.]	O-CONT.	@-TO-ω(L)	LIN.[16]	NO REC.
a. [$_{\omega^{max}}$ simato]	*!				
b. [$_{\omega^{max}}$ masito]		*!		*	
☞ c. [$_{\omega^{max}}$ matosi]				*	
d. [$_{\omega^{max}}$ [$_{\omega}$ mato] si]				*	*!

Candidate a does not abide by high-ranking IDENT[antitropal], and hence is discarded. Candidate b abides by this constraint but violates the constraint against infixation, O-CONTIGUITY. Candidate c, with a *si* displaced to the rightmost position, wins out, as it does not involve infixation, although it violates low-ranking LINEARITY. This candidate does not show any iteration of prosodic levels. Candidate d, by contrast, introduces a boundary between the verbal base and *si*, which induces a violation of NO RECURSION. As will be shown in subsection 7.2, segmental effects indicate that candidate c, and not candidate d, is indeed the right candidate.

We deal next with a prefixed verbal form, *pasimato* ‘(they) see each other’, the input to prosody being pa-[*si*-mat-o].

(79) *Derivation of forms with one prefix*

	pa-[*si*-mat-o]	ID[ant.]	O-CONT.	@-TO-ω(L)	LIN.	NO REC.
☞	a. [$_{\omega^{max}}$ pa [$_{\omega}$ simato]]					*
	b. [$_{\omega^{max}}$ pasimato]			*!		
	c. [$_{\omega^{max}}$ pa [$_{\omega}$ matosi]]				*!	*

Candidate a is the winner. It trivially abides by IDENT[antitropal], since *si*, remaining in situ, is not aligned with an ω^{max}-boundary. It respects all other constraints except for NO RECURSION, as the sequence *simato* is parsed twice as a prosodic word. Candidate c is almost identical to candidate a, but it loses out because it shows an unwarrantedly displaced *si*, violating LINEARITY. Finally, candidate b violates the higher ranking constraint @-TO-ω(L), as the left edge of the linearized @-span is not aligned with the left edge of the prosodic word. That there is an ω-boundary between the LP and *si* is revealed by segmental effects described in section 7.2.

Any added prefix does not modify this last scenario at all. For instance, inputting ne-pa-[*si*-mat-o] '(they) don't see each other' also gives back a winning candidate where *si* appears immediately before the verbal base, after all prefixes, and where each of the prefixes is parsed as iterating the ω-level.

(80) *Derivation of forms with more than one prefix*

	ne-pa-[*si*-mat-o]	ID[ant.]	O-CONT.	@-TO-ω(L)	LIN.	NO REC.
☞	a. [$_{\omega^{max}}$ ne [$_{\omega}$ pa [$_{\omega}$ simato]]]					**
	b. [$_{\omega^{max}}$ nepasimato]			*!		
	c. [$_{\omega^{max}}$ ne [$_{\omega}$ pa [$_{\omega}$ matosi]]]				*!	**
	d. [$_{\omega^{max}}$ ne [$_{\omega}$ si [$_{\omega}$ pa [$_{\omega}$ mato]]]]				*!	***

Another OT-approach to the position of Lithuanian *si* appeared in Kushnir 2025, an article published in April 2025, while our paper was undergoing a first round of revisions.[17] Similarly to us, Kushnir assumes that *si* realizes a Voice head in the syntax, that its default position is as a prefix adjacent to the verbal base, and that a prosodic constraint enforces its suffixation as the rightmost exponent when no other prefix is present. However, Kushnir's approach can be said to be less parsimonious and explanatory than ours. Kushnir adopts a lexicalist analysis where syntax and morphology are two separate modules, the mapping between the two regulated by a set of Merge Condition constraints, one for each syntactic formative. For instance, MCVoice ensures that the merger of the syntactic head Voice will optimally trigger the addition of an exponent in the morphology. As regards the position of the exponents within the word, it is enforced via alignment constraints that seem to be, unlike in our analysis, specific to Lithuanian, like the one ensuring the suffixation of Agreement or the one ensuring the prefixation of the negation. By contrast, in our approach, with the exception of the lexical specification of the default prefixal linearization of *si*, the position of the rest of exponents follows from either a default suffixation mechanism (as in the case of the tense+agreement exponent) or the fact that they correspond to spans of their own in the syntax (the case of the lexical and the superlexical prefixes), as backed by independent syntactic-semantic evidence (see footnote 10). In addition, the constraints used in our OT-analysis refer to universal conditions that operate similarly in other languages, capturing the behavior of other second position elements like clitic pronouns in EP. Finally, while most of Kushnir's alignment constraints make reference directly to the syntactic heads (rather than exponents), one of them refers to "preverbs", which, as Kushnir himself points out, relates to two different positions in the syntactic configuration (Asp and Appl), adding a further stipulation to the syntax-

morphology mapping.

This section has shown that by taking the initial linearization to be as in (25), we can formulate the displacement of *si* as a true second position effect, or more accurately, as a non-first-position effect.[18] We derive this effect from a prosodically driven edge-avoidance constraint of the ilk of those found in many languages (Halpern and Zwicky 1996, Bye and Svenonius 2012). Our approach also challenges the analyses using pre-prosodic operations such as Local Dislocation (Embick and Noyer 2001). Additionally, if our analysis is on the right track, it provides new evidence for recursion in prosody, by showing that the maximal projections of segments can be referred to by constraints (Ito and Mester 2009). The next section presents predictions made by this prosodic account and unavailable to previous ones.

7 Predictions for Morphology and Phonology

7.1 Insensitivity of si-*Suffixation to Morphological Boundaries*

The mechanism that we propose to account for the suffixal position of *si* is expressed in terms of prosodic constituents and is only active at the prosodic level. Therefore, it is predicted to disregard morphological boundaries. As such, when suffixed, *si* will always be the rightmost exponent of the prosodic word, regardless of any morphological boundaries. For verbs, this includes overt verbalizers (81) and all verbal inflection (82).

(81) aug-**in**-a-si (cf. *augsiina)
grow-V-3.PRS-*si*
'grows (sth) for self'

(82) mat-**o**-si (cf. *matsio)
see-3.PRES-*si*
'is/are visible'

But *si* also skips material related to any other category when the verb serves as the base for derivation, like nominalizers (83) and participial morphology (84), together with their inflectional material.

(83) mat]$_{v}$-**ym**-**as**]$_{n}$-is (cf. *matsiymas)
see-NMLZ-NOM.M.SG-*si*
'the seeing of each other'

(84) mok]$_{v}$-**ęs**]$_{ptcp}$-is (cf. *moksięs)
learn-PTCP.PST.NOM.M.SG-*si*
'learnt'

In all the above cases, the position of *si* is correctly predicted in our account. This is because the base to which *si* is suffixed is a (minimal) prosodic word, which is, in turn, what is predicted to be mapped from an @-span in the syntax. And indeed, all the bases in the examples above count as @-spans in the syntax. To illustrate, consider the derivation of the nominalization *aug-in-im-as-is* 'the growing of something for oneself', assuming a pretty standard syntactic configuration, where, as expected within the Spanning approach, every single feature corresponds to a head.

(85) $\langle K^{w} > Num > Gen > n^{@} > Voice_{refl}{}^{w} > v > \sqrt{AUG} \rangle$

The verbal layer is embedded under the nominal layer, up to Voice, and the nominal layer exhibits the functional heads Gender (masculine), Number (singular), and Case (K: nominative). The nominalization involves two cycles of lexical access, related to each of the two categories involved (Embick 2010). This is marked with the w dia-

critic on the highest verbal functional head, Voice, and the highest nominal functional head, K. The exponents spelling out these spans are as follows:

(86) /aug/ ⇔ $\langle\sqrt{\text{AUG}}\rangle$ /im/ ⇔ ⟨n⟩

/in/ ⇔ ⟨v⟩ / {$\sqrt{\text{AUG}}$, etc.} __ /as/ ⇔ ⟨Nom > Sg > Masc⟩

si- ⇔ $\langle\text{Voice}_{\text{refl}}\rangle$

All exponents are suffixed to the lowermost one, the root, except for *si*, which, as described in section 5, is prefixed. There is a single @ diacritic, on the nominal categorizer. The whole span counts as an @-span. The output of the linearization algorithm is shown in (87).

(87) [*si*-aug-in-im-as]

Given the ranking of constraints assumed, the prediction for the prosodification of this sequence is exactly the same as that for non-prefixed verbal form (cf. (78)), that is, *si* will optimally appear at the right boundary of its prosodic word: *auginimas**is***.

Things are different, of course, if there is any additional prefix, like the lexical prefix *už*. In that case, *už* being merged as an independent span (a specifier), the following representation is delivered by the linearization algorithm:

(88) už-[*si*-aug-in-im-as]

‘the (successful) growing of something for oneself’

In this case, the optimal prosodic representation is one in which *si* appears right adjacent to *už* and left-adjacent to the minimal prosodic word, since its antitropal diacritic is abided by. Crucially, even though in both (87) and (88) *si* is at the left edge of a linearized @-span, it is only in (87) that it is at the left edge of an ω^{max}, which is what

the antitropal diacritic is sensitive to.

In sum, the fact that *si*, in the absence of other prefixes, ends up suffixed to the rightmost boundary of the word, no matter the morphological cycles contained, comes for free with the prosodic account, and accounts that are not based on prosody cannot explain this state of affairs with as simple a mechanism.

7.2 The Prefix/Suffix Asymmetry as Revealed by Segmental Interactions

The second set of predictions has to do with certain phonological effects triggered when *si* is a suffix. According to our account, *si* is always associated with the same prosodic word: the one generated from the @-span of which $\langle \text{Voice}_{\text{refl}} \rangle$ is itself a subspan. *Si* is a prefix when located on its left boundary, and it is a suffix when located on its right boundary. It is well-known that prefixes and suffixes show different phonological behavior, crosslinguistically (see, for instance, Elkins 2020). Furthermore, we predict that there is an ω-boundary between any prefix and the reflexive.[19] We therefore expect different phonological interactions when *si* is parsed as a prefix and as a suffix. As we will show in this section, these predictions are borne out.

When *si* is a suffix to the base, we see an array of phonological interactions between the final phoneme of the base and *si*. A prominent class of such interactions involve bases ending in long vowels or diphthongs. (89) shows a second person plural verb form. When followed by *si* in (89a), the quality of the verbal base vowel is [e:], and the reflexive is /s/.[20] When the verb occurs without *si*, the base ends in [ɛ]. We analyse this as the clipping of an underlying /e:/ at the right ω-boundary (see Kenstowicz 1971:53).

(89) a. mat-ot[e:]-s ‘you are visible’
see-PRS.2PL-*si*

b. mat-ot[ɛ] ‘you see’
see-PRS.2PL

Similar clipping is observed with underlying diphthongs. A base ending in /uɔ/ or /iɛ/ preserves the diphthongs with postposed *si*, as shown in (90a) and (91a). In the absence of *si*, the diphthong is monophthongized (clipped) to [ʊ] and [ɪ] respectively, as it occurs at the ω-boundary—see (90b) and (91b).[21]

(90) a. keli-[uɔ]-si
raise-PRS.1SG-*si*
‘I get up, arise’

b. keli-[ʊ]
raise-PRS.1SG
‘I raise, lift up’

(91) a. kel-[iɛ]-si
raise-PRS.2SG-*si*
‘you get up, arise’

b. kel-[ɪ]
raise-PRS.2SG
‘you raise, lift up’

Another interaction takes place when the postposed reflexive follows a consonant. In this case, metathesis occurs to dissimilate (Ambrazas et al. 1997), yielding *is*.

(92) riš-**is**
tie.FUT.3-*si*
‘will tie (something for self)’

(93) mok-ant-**is**
teach-CVB.PRS-*si*
‘(while) learning’

All these interactions are unattested between preposed *si* and the segments of a preceding prefix, even in the same segmental environments. Thus, consonant clusters are not dissimilated.[22]

(94) a. iš-**si**-keli-a
LP-*si*-raise-PRS.3
‘moves out’

b. *iš-is-keli-a

Similarly, preposed *si* does not preserve the diphthong of a preceding prefix. Indeed, some LPs can be shown to underlyingly involve a diphthong. Thus, *nu-* and *pri-* have diphthongal variants in certain simple, non-eventive nouns: ***nuo**-lauža* 'broken-apart piece, splinter', ***prie**-statas* 'annex'. Unlike the syntax described for LPs in eventive, resultative expressions, for these cases, we take the prefixal root ($\sqrt{\text{NUO}}$, $\sqrt{\text{PRIE}}$) to merge directly with the verbal root ($\sqrt{\text{LAUŽ}}$, $\sqrt{\text{STAT}}$): there is no ResP projecting, nor v, which accounts for the non-eventive semantics of these nouns. Both roots are thus embedded under nominal structure, within the same @-span, hence the same ω^{min}. Since the diphthong is not aligned with an ω-boundary, it is preserved. By contrast, in the cognate resultative verbs the prefix corresponds to a ResP, which constitutes an independent span and thus initiates an independent lexical cycle. Thus, an ω-boundary separates the prefix and the base, which triggers the clipping of the underlying diphthong: ***nu**-laužti* 'to break', ***pri**-statyti* 'to build (sth) next to (sth), to annex'. As expected, forms involving *si* also show a clipped prefix, as there is an ω-boundary between the prefix and the reflexive.

(95) a. n[ʊ]/*n[uo]-si-mato
LP-*si*-see.PRS.3
'(it) is foreseable'

b. pr[ɪ]/*pr[iɛ]-si-mano
LP-*si*-think.PRS.3
'(he/she) imagines'

The state of affairs described in this section is not the one predicted by approaches like Embick and Noyer's (2001), Bruening's (2019), or Stump's (2022), in which *si* is not always associated to the base.[23] For instance, in order to account for the puzzling linearization of *si*, Stump (2022) proposes that *si* is always a suffix. It is a suffix to the inflected verb when appearing after it, but it is also a suffix when appearing after all prefixes, that is, it is a suffix to those prefixes. Stump's (2022) characterization thus predicts *si* to interact with a preceding prefix in the same way that it does with the

verbal base, contrary to fact.

8 Conclusions

Our account of the classic case study of the Lithuanian reflexive *si* derives its linearization from the syntax using Spanning, and explains the displacement of *si* prosodically, doing away with idiosyncratic morphological movement.

We have shown that previous explanations of the position of *si*, either syntactic (Korostenskienė 2014, 2016a, 2016b, Bruening 2019) or morphological (Embick and Noyer 2001, Embick 2007, Arregi and Nevins 2012, Šereikaitė 2017, Adamson 2019, Stump 2022), have been inadequate.

We have proposed that *si* is an exponent of a Voice head, which linearizes before the base.

(25) (prefixes-)*si*-inflected verbal base

Syntax suggests its initial linearization, but prosody has the final word: if *si* appears at the left edge of a maximal prosodic word, it has to escape this environment by suffixing to the right of the right-adjacent prosodic word. Hence, this is a true Wackernagel (second-position) phenomenon (conceived as first-position avoidance), but it is determined prosodically, and not morphologically, like previous proposals suggested. In addition, our account easily captures the insensitivity of suffixal *si* to morphological cycles and the facts about the phonological interactions between *si* and its host.

By showing that no morphological explanation is available, our account directly opposes those that use post-syntactic operations (Local Dislocation: Embick and Noyer 2001; Metathesis: Arregi and Nevins 2012) in DM. We do away with idiosyncratic instances of post-syntactic movement, and instead attribute the phenomenon to the

prosodic component (see also Kushnir 2025, who, however, adds a morphological movement rule to effect suffixation). This is crosslinguistically supported. We provide a direct comparison of two only distantly related Indo-European languages, Portuguese and Lithuanian: the alignment with the left edge of a prosodic constituent is identical in both, only the constituent size differs.

We show that syntactic elements can be sensitive to prosody, tightening the connection between syntax and phonology, and providing support to the theories of linearization such as Spanning in which syntactic relations map in a nontrivial way to prosodic representations. Our account also provides novel evidence for prosodic recursion as built on syntactic recursion and suggests that languages can exhibit prosodic phenomena that refer to the maximal segment of a prosodic constituent.

References

Acedo-Matellán, Víctor. 2010. Argument structure and the syntax-morphology interface. A case study in Latin and other languages. Doctoral dissertation, Universitat de Barcelona.

Acedo-Matellán, Víctor. 2016. *The morphosyntax of transitions: A case study in Latin and other languages*. Oxford University Press.

Acedo-Matellán, Víctor, and Arkadiusz Kwapiszewski. 2026. Talmy's typology revisited: A spanning approach. *Linguistic Inquiry* 57:537–563. https://doi.org/10.1162/ling_a_00531.

Adamson, Luke J. 2019. Derivational trapping and the morphosyntax of inflectionlessness. Doctoral dissertation, University of Pennsylvania.

Ambrazas, Vytautas, Emma Geniušienė, Aleksas Girdenis, Nijolė Sližienė, Dalija Tekorienė, Adelė Valeckienė, and Elena Valiulytė. 1997. *Lithuanian grammar*. Baltos lankos.

Arkadiev, Peter. 2011. On the aspectual uses of the prefix *be-* in Lithuanian. *Baltic linguistics* 2:37–78.

Arkadiev, Peter. 2012. "External prefixes" in Lithuanian: A typological rarum. Paper presented at Baltic Languages and White Nights, Helsinki, 11-12 June 2012.

Arregi, Karlos, and Andrew Nevins. 2012. *Morphotactics: Basque auxiliaries and the structure of spellout*. Springer.

Asyllogistou, Athanasia. 2019. Result in the diachrony of Greek. Doctoral dissertation, Ulster University.

Barbosa, Pilar. 1996. Clitic placement in European Portuguese and the position of

subjects. In *Approaching second: Second position clitics and related phenomena*, ed. by Aaron L. Halpern and Arnold M. Zwicky, 1–40. CSLI Publications.

Bennett, Ryan, Emily Elfner, and James McCloskey. 2016. Lightest to the Right: An Apparently Anomalous Displacement in Irish. *Linguistic Inquiry* 47:169–234. https://doi.org/10.1162/LING_a_00209.

Billings, Loren A. 2002. Phrasal clitics. *Journal of Slavic Linguistics* 10:53–104.

Booij, Geert E. 1983. Principles and parameters in prosodic phonology. *Linguistics* 21:249–280.

Bošković, Željko. 2001. *On the nature of the syntax-phonology interface: Cliticization and related phenomena*. Elsevier.

Brody, Michael. 2000. Mirror theory: Syntactic representation in perfect syntax. *Linguistic Inquiry* 31:29–56.

Bruening, Benjamin. 2013. *By* phrases in passives and nominals. *Syntax* 16:1–41.

Bruening, Benjamin. 2019. Doing away with post-syntactic operations: No need for lowering or local dislocation in a minimal morphosyntax. Ms., University of Delaware.

Bye, Patrik, and Peter Svenonius. 2012. Non-concatenative morphology as epiphenomenon. In *The morphology and phonology of exponence*, ed. by Jochen Trommer, 427–495. Oxford University Press.

Chomsky, Noam. 2000. Minimalist inquiries: The framework. In *Step by step: Essays on minimalist syntax in honor of Howard Lasnik*, ed. by Roger Martin, David Michaels, Juan Uriagereka, and Samuel J. Keyser, 89–155. MIT Press.

Chomsky, Noam. 2001. Derivation by phase. In *Ken Hale: A life in language*, ed. by Michael Kenstowicz, 1–52. MIT Press.

Chung, Sandra. 2003. The syntax and prosody of weak pronouns in Chamorro. *Linguistic Inquiry* 34:547–599. https://doi.org/10.1162/002438903322520151.

Elkins, Noah E. 2020. Prefix independence: typology and theory. Doctoral dissertation, University of California, Los Angeles.

Embick, David. 2004. On the structure of resultative participles in English. *Linguistic Inquiry* 35:355–392.

Embick, David. 2007. Linearization and Local Dislocation: Derivational mechanics and interactions. *Linguistic Analysis* 33:303–336.

Embick, David. 2010. *Localism versus globalism in morphology and phonology*. MIT Press.

Embick, David, and Rolf Noyer. 2001. Movement operations after syntax. *Linguistic Inquiry* 32:555–595. https://doi.org/10.1162/002438901753373005.

Evans, Nick. 1995. A-quantifiers and scope in Mayali. In *Quantification in natural languages*, ed. by Emmon Bach, Eloise Jelinek, Angelika Kratzer, and Barbara H. Partee, 207–270. Kluwer.

Folli, Raffaella, and Heidi Harley. 2020. A head movement approach to Talmy's typology. *Linguistic Inquiry* 51:425–470.

Franks, Steven, and Tracy Holloway King. 2000. *A handbook of Slavic clitics*. Oxford Studies in Comparative Syntax. Oxford University Press.

Garrett, Andrew. 1996. Wackernagel's law and unaccusativity in Hittite. In *Approaching second: Second position clitics and related phenomena*, ed. by Aaron L. Halpern and Arnold M. Zwicky, 85–133. CSLI Publications.

Geniušienė, Emma. 1987. *The typology of reflexives*. Walter de Gruyter.

Hale, Mark. 1996. Deriving Wackernagel's law: Prosodic and syntactic factors determining clitic placement in the language of the Rigveda. In *Approaching sec-*

ond: Second position clitics and related phenomena, ed. by Aaron L. Halpern and Arnold M. Zwicky, 165–197. CSLI Publications.

Halpern, A. L. 1992. Topics in the placement and morphology of clitics. Doctoral dissertation, Stanford University.

Halpern, Aaron L., and Arnold M. Zwicky. 1996. *Approaching second: second position clitics and related phenomena*. CSLI Publications.

Haselbach, Boris, and Marcel Pitteroff. 2015. A morphological case approach to PPs. *Linguistische Berichte* 244:425–460.

Herd, Jonathon. 2003. Deriving prosodic inversion: Clitics, cyclicity and the organization of post-syntactic interfaces. *Toronto Working Papers in Linguistics* 21.

Hock, Hans Henrich. 1996. Who's on first? Toward a prosodic account of P2 clitics. In *Approaching second: Second position clitics and related phenomena*, ed. by Aaron L. Halpern and Arnold M. Zwicky, 199–270. CSLI Publications.

Hoekstra, Teun, and René Mulder. 1990. Unergatives as copular verbs: Locational and existential predication. *The Linguistic Review* 7:179.

Ito, Junko, and Armin Mester. 2009. The extended prosodic word. In *Phonological domains: Universals and deviations*, ed. by Janet Grijzenhout and Baris Kabak, 135–194. Mouton de Gruyter.

Jakaitienė, Evalda, Adelė Laigonaitė, and Aldona Paulauskienė. 1976. *Lietuvių kalbos morfologija* [*Morphology of the Lithuanian language*]. Mokslas.

Kastner, Itamar. 2020. *Voice at the interfaces: The syntax, semantics, and morphology of the Hebrew verb*. Language Science Press.

Kayne, Richard S. 1975. *French syntax: The transformational cycle*. MIT Press.

Kenstowicz, Michael John. 1971. Lithuanian phonology. Doctoral dissertation, University of Illinois at Urbana-Champaign.

Korostenskaja, Julija. 2014. Setting the verb raising parameter in Lithuanian. *Kalbų studijos* 25:14–25.

Korostenskienė, Julija. 2014. Exploring Lithuanian reflexive verbs from the generative syntactic perspective. *Deeds and Days / Darbai ir Dienos* 62:59–80.

Korostenskienė, Julija. 2015. On some properties of Lithuanian verbal prefixes. *Verbum* 6:95–122. https://doi.org/10.15388/Verb.2015.6.8811.

Korostenskienė, Julija. 2016a. On the placement of the reflexive-middle marker *-si-/-sja* in Lithuanian and Russian. *Language and Contexts* 7:61–72. https://doi.org/10.5755/j01.sal.0.28.15126.

Korostenskienė, Julija. 2016b. On the placement of the reflexive/reciprocal marker *-si-* in Lithuanian verbs. *Kalbų studijos/Studies about Language* 28:34–51. https://doi.org/10.5755/j01.sal.0.28.15126.

Korostenskienė, Julija. 2017. On binding, lexical and superlexical prefixes, and *si* in the Baltic verb. *Canadian Journal of Linguistics/Revue canadienne de linguistique* 62:449–501.

Korostenskienė, Julija. 2022. Baltic topmost superlexical prefixes as the left periphery of the verb: The permissive, the restrictive, the negative, and the debitive. *Journal of Baltic Studies* 53:1–27. https://doi.org/10.1080/01629778.2022.2040554.

Kozhanov, Kirill. 2016. Verbal prefixation and argument structure in Lithuanian. In *Argument realization in Baltic*, ed. by Axel Holvoet and Nicole Nau, 363–402. John Benjamins. https://doi.org/10.1075/vargreb.3.

Kushnir, Yuriy. 2025. The variable position of the reflexive affix in Lithuanian. *Natural Language & Linguistic Theory* 43:2257–2273.

Kwapiszewski, Arkadiusz. 2022. Aspect and event structure: the morphosyntax of

Polish verbs from a cross-linguistic perspective. Doctoral dissertation, University of Oxford. https://doi.org/10.5287/ora-zrqdj2xgb.

Labelle, Marie. 2008. The French reflexive and reciprocal *se*. *Natural Language & Linguistic Theory* 26:833–876. https://doi.org/10.1007/s11049-008-9053-1.

Marelj, Marijana, and Eric J. Reuland. 2013. Clitic SE in Romance and Slavonic revisited. In *Current studies in slavic linguistics*, ed. by Irina Kor Chahine, volume 146 of *Studies in Language. Companion Series*, 75–88. John Benjamins.

Mateu, Jaume. 2008. On the l-syntax of directionality/resultativity: The case of Germanic preverbs. In *Syntax and semantics of spatial P*, ed. by Anna Asbury, Jakub Dotlačil, Berit Gehrke, and Rick Nouwen, 221–250. John Benjamins. https://doi.org/10.1075/la.120.

McGinnis, Martha. 2022. Reflexive clitics are verbal, not pronominal. *Canadian Journal of Linguistics/Revue canadienne de linguistique* 67:328–352. https://doi.org/10.1017/cnj.2016.38.

Nespor, Marina, and Irene Vogel. [1982] 2012. *Prosodic phonology: With a new foreword (2nd ed. [with a new preface])*. De Gruyter Mouton. https://doi.org/10.1515/9783110977790.

Nevis, Joel A., and Brian D. Joseph. 1993. Wackernagel affixes: Evidence from Balto-Slavic. In *Yearbook of morphology 1992*, ed. by Geert Booij and Jaap van Marle, 93–111. Springer Netherlands. https://doi.org/10.1007/978-94-017-3710-4_4.

Pesetsky, David. 1995. *Zero syntax: Experiencers and cascades*. MIT Press.

Pollock, Jean-Yves. 1989. Verb movement, universal grammar, and the structure of IP. *Linguistic Inquiry* 20:365–424.

Pylkkänen, Liina. 2008. *Introducing arguments*. MIT press. https://doi.org/10.7551/mitpress/9780262162548.001.0001.

Radanović-Kocić, Vesna. 1988. The grammar of Serbo-Croatian clitics: A synchronic and diachronic perspective. Doctoral dissertation, University of Illinois at Urbana-Champaign.

Radanović-Kocić, Vesna. 1996. A prosodic account of clitic position in Ancient Greek. In *The placement of Serbo-Croatian clitics: A prosodic approach*, ed. by Aaron L. Halpern and Arnold M. Zwicky, 429–445. CSLI Publications.

Ramchand, Gillian. 2008. *Verb meaning and the lexicon: A first phase syntax*. Cambridge University Press. https://doi.org/10.1017/CBO9780511486319.

Ruskan, Anna. 2019. Functional variation of discourse particles in Lithuanian: A look at clause peripheries. *Corpus Pragmatics* 3:303–325.

Schäfer, Florian. 2008. *The syntax of (anti-)causatives*. John Benjamins. https://doi.org/10.1075/la.126.

Šereikaitė, Milena. 2017. The reanalysis of Lithuanian reflexive -si-: a DM approach. In *Proceedings of the 34th West Coast Conference on Formal Linguistics*, ed. by Aaron Kaplan, Abby Kaplan, Miranda K. McCarvel, and Edward J. Rubin, 447–453. Cascadilla Proceedings Project.

Šereikaitė, Milena. 2018. Lexical vs. nominal prefixes and their meaning domains. In *University of Pennsylvania Working Papers in Linguistics*, volume 24. University of Pennsylvania.

Šereikaitė, Milena. 2020. Voice and case phenomena in Lithuanian morphosyntax. Doctoral dissertation, University of Pennsylvania.

Šereikaitė, Milena. 2022. The dual nature of Lithuanian reflexive *-si-*: remarks on Lo-

cal Dislocation Talk given at the Princeton Phonology Forum, Princeton University, December 2.

Šereikaitė, Milena. 2023. Movement in mobile affixation: A view from Balto-Slavic Talk given at Rutgers University, April 28.

Šereikaitė, Milena. 2026. Case and voice properties of complex event nominalizations: A voice-bundling approach. *Natural Language & Linguistic Theory* 44. https://doi.org/10.1007/s11049-025-09685-1.

Sigurðsson, Einar Freyr, and Milena Šereikaitė. 2024. The dual face of structural object case: On lithuanian genitive of negation. *Journal of Linguistics* 60:161–212.

Starke, Michal. 2018. Complex left branches, spellout, and prefixes. In *Exploring Nanosyntax*, ed. by Lena Baunaz, Liliane Haegeman, Karen De Clercq, and Eric Lander. Oxford University Press. https://doi.org/10.1093/oso/9780190876746.003.0009.

Stump, Gregory. 2022. *Morphotactics: A rule-combining approach*. Cambridge University Press. https://doi.org/10.1017/9781009168205.

Svenonius, Peter. 2004. Slavic prefixes inside and outside VP. *Nordlyd* 32:205–253. https://doi.org/10.7557/12.68.

Svenonius, Peter. 2008. Russian prefixes are phrasal. In *Formal description of Slavic languages*, ed. by Gerhild Zybatow, Luka Szucsich, Uwe Junghanns, and Roland Meyer, 526– 537. Peter Lang.

Svenonius, Peter. 2016. Spans and words. In *Morphological metatheory*, ed. by Heidi Harley and Daniel Siddiqi, 199–220. John Benjamins.

Tan, Tamisha L. 2024. The morphosyntax of Gothic preverb compounds: Incorporation and applicativisation. *The Journal of Comparative Germanic Linguistics* 27. https://doi.org/10.1007/s10828-024-09151-z.

Taylor, Ann. 1996. A prosodic account of clitic position in Ancient Greek. In *Approaching second: Second position clitics and related phenomena*, ed. by Aaron L. Halpern and Arnold M. Zwicky, 477–503. CSLI Publications.

Tolskaya, Inna. 2018. Nanosyntax of Russian verbal prefixes. In *Exploring Nanosyntax*, ed. by Lena Baunaz, Karen De Clercq, Liliane Haegeman, and Eric Lander, 205–236. Oxford University Press. https://doi.org/10.1093/oso/9780190876746.003.0008.

Ulvydas, Kazys. 1971. *Lietuvių kalbos gramatika. Tomas I. Morfologija.* [*A grammar of the Lithuanian language. Vol I. Morphology*]. Mintis.

Utka, Andrius, Erika Rimkutė, Jolanta Kovalevskaitė, Agnė Bielinskienė, Mažvydas Petkevičius, Rūta Petrauskaitė, and Jurgita Mikelionienė. 2017. *Dabartinės lietuvių kalbos tekstynas* [*Corpus of the Contemporary Lithuanian Language]*. http://hdl.handle.net/20.500.11821/16, CLARIN-LT digital library in the Republic of Lithuania.

Wackernagel, Jacob. 1892. Über ein Gesetz der indogermanischen Wortstellung. *Indogermanische Forschungen* 1:333–436.

Williams, Edwin. 2002. *Representation Theory*. MIT Press. https://doi.org/10.7551/mitpress/5893.001.0001.

Williams, Edwin. 2007. Dumping lexicalism. In *The Oxford Handbook of Linguistic Interfaces*, ed. by Gillian Ramchand and Charles Reiss, 353—382. Oxford University Press. https://doi.org/10.1093/oxfordhb/9780199247455.013.0012.

Wood, Jim. 2015. *Icelandic morphosyntax and argument structure*. Springer. https://doi.org/10.1007/978-3-319-09138-9.

Wood, Jim, and Alec Marantz. 2017. The interpretation of external arguments. In *The verbal domain*, ed. by Roberta D'Alessandro, Irene Franco, and Ángel J. Gal-

lego, 255–278. Oxford University Press. https://doi.org/10.1093/oso/9780198767886.003.0011.

Zwicky, Arnold M. 1985. Clitics and particles. *Language* 61:283–305.

Víctor Acedo-Matellán

Faculty of Linguistics, Philology, and Phonetics

University of Oxford

victor.acedo-matellan@oriel.ox.ac.uk

Veronika Gvozdovaitė

Department of Linguistics

New York University

veronika.g@nyu.edu

Notes

We are grateful to two anonymous reviewers whose comments and suggestions have greatly improved the quality of this paper. Versions of this research were presented at ConSOLE 32 and at seminars at the University of Oxford, the University of Cambridge, Universitat de Barcelona, and Universitat Rovira i Virgili. We thank the audiences at these events for their insightful questions and observations. Special thanks go to Maria Gouskova and Francesc Torres-Tamarit. Any remaining shortcomings are our own responsibility.

[1]For previous non-generative approaches see Ulvydas 1971, Jakaitienė et al. 1976, Geniušienė 1987, and Ambrazas et al. 1997.

[2]Negative *ne-*, modal *te-* and aspectual *be-* prefixes, sometimes also called *external prefixes* (Arkadiev 2012), have been argued to be "superlexical": they can stack on top of lexical prefixes, they do not change the argument structure properties of the verb, and they do not induce special meanings in the verb (see Korostenskienė 2015 for Lithuanian). However, they still differ from the prefixes that are termed *superlexical* in the Slavic literature (Svenonius 2004 and subsequent): they do not have quantizing meanings, at least *ne-* and *te-* are not of prepositional origin, and they can excorporate. It is of note that *be-* usually occurs together with *te-* or *ne-*, and so *tebe-* and *nebe-* are sometimes treated as a single complex superlexical prefix. We maintain the morphemic boundary mark in *te-be-* and *ne-be-* on the grounds of each prefix occurring on its own and each contributing predictable (compositional) meaning components. However, treating them as a complex prefix does not affect our argument. See also footnote 10.

[3]*Si* may surface as *is* or *s* when suffixed. These alternative forms are dealt with in subsection 7.2.

[4]A further question is whether the verb is linearized at Voice or lower. Korostenskaja

(2014) argues that the verb is linearized in its base position, based on the fact that manner adverbs like 'completely' precede the verb. This seems consistent with the fact that Voice-level (agentive) adverbs such as *tyčia* 'on purpose' are linearized before the verb.

(i) Marija dažnai tyčia dainuoja.
Marija often on_purpose sings
'Marija often sings on purpose.'

However, we observe that non-agentive adverbs like *netyčia* 'by accident', not used by Korostenskaja (2014), can be neutrally linearized after the inflected verb, as shown in (ii).

(ii) Marija nukrito netyčia.
Marija fell accidentally
'Marija fell by accident.'

Non-agentive adverbs thus contrast with other adverbs, which can only appear postverbally when focused (in (iii)), but are otherwise preverbal with completely neutral intonation (wide focus, in (i)).

(iii) Marija dainuoja dažnai (o ne retai).
Marija sings often and not rarely
'Marija sings often (and not rarely).'

The fact that all adverbs are linearized before the verb, except for the non-agentive *netyčia* 'by accident', which is neutrally linearized after the verb, suggest that the Lithuanian verb stays low, but not in its base position. Under the assumption that all predicates involve a Voice head, thematic (in non-unaccusatives) or not (in unaccusatives) (see, for instance, Wood 2015 and section 3.2), the empirical facts described can be explained by the hypothesis, which we will adopt, that the verb linearizes at Voice: non-agentive adverbs like *netyčia* attach to vP, that is, the verb's base position, and all other adverbs attach to either Voice

(thematic or not) or higher. We will be assuming this.

A full investigation of the issue, however, would include a more comprehensive discussion of unaccusativity in Lithuanian, including devising diagnostics, looking into the scopal properties of post-verbal *netyčia* 'accidentally' (which might end up liking it to focused adverbs), and investigating the cooccurrence restrictions on this kind of adverb (*netyčia* 'accidentally' seems to rarely occur with manner adverbs). This endeavor is beyond the scope of the present paper. What seems to be proven, and is relevant for the discussion in this section, is that the verb never raises to Asp.

[5]An anonymous reviewer brings our attention to the verb *į-si-*[[*pa-vidal*]$_{\text{N}}$*-in*]$_{\text{V}}$*-ti* (the reviewer's decomposition) 'to acquire a form', where similarly to the two verbs we have discussed, *si* seems to ignore the *pa* adjacent to the root. We observe that, in this case, there is even less evidence for the synchronic status of *pa-* as a prefix rather than as part of a root *pavidal*, since *vidal* is not found in the absence of *pa* in Lithuanian.

The verb *parduoti* 'to sell' behaves yet differently. As such, it shows a regular pattern, with *si* intervening between the LP *par-* and the base *duo* 'give', as in the infinitive *par-si-duoti* 'to sell oneself'. However, this verb admits the use of other LPs, like *iš-*. When a further LP is used, *si* appears between both LPs: *iš-si-par-duoti* 'to sell out'. Interestingly, *par-* is a productive prefix (*par-kristi* 'to fall down', *par-nešti* 'to bring back', *par-keliauti* 'to travel back', etc.), but only *parduoti* occurs with a second LP, which suggests that *parduo* may be available as a single root, making linearizations like *iš-si-parduoti* 'to sell out' regular. This is in line with Kostenskienė's (2014) proposal that *par-* may incorporate with the root under the pressure of another LP.

[6]An anonymous reviewer observes, illustrating with the case in (i) (from Šereikaitė 2026:ex. (25)), that agent nominalizations may also feature *si*.

(i) iš-si-suk-inė-toj-as
PFV-REFL-turn-IPFV-AGN-NOM.M.SG
'a shuffler/shirk'

If Šereikaitė (2020, 2026) is correct that agent nominalizations do not contain Voice, the presence of *si* in examples like (i) would in principle contradict our analysis. While space limitations preclude a full exploration of this issue, which we leave for future research, we would like to offer some reflections. First, it could be the case that agent nominalizations are indeed compatible with the reflexive/anticausative Voice head that we are entertaining in this section, and incompatible with other kinds of Voice. We note, in addition, that the presence of the imperfective suffix *-inė* in agent nominalizations like (i) would pose a problem for the claim that such nominalizations cannot contain Voice, if it is the case that *-inė* represents viewpoint aspect, by hypothesis merged above Voice, rather than inner aspect. In that sense, it is significant that, as pointed out by Korostenskienė (2022), aspectual suffixes like *-inė* scope over the LP-prefixed verb.

[7]One of our anonymous reviewers actually invited us to emphasize that Lithuanian *si* is not a clitic. See also McGinnis 2022 for the broader claim that reflexive "clitics" are the realization of Voice heads, rather than pronouns.

[8]An anonymous reviewer observes that *si* is compatible with periphrastic passives, the "passiveness" residing in the participial suffix *-t*.

(i) Šis paveikslas yra (jų)
this.NOM.SG picture.NOM.SG be.PRS.3 they.GEN
at-si-vež-t-as iš Ispanijos.
LP-*si*-take-PTCP.PST.PASS-NOM.M.SG from Spain.GEN
'This picture was brought (by them) from Spain.'

We argue that such examples do not jeopardize the analysis of *si* as a Voice head. Rather than assuming that the periphrastic passive involves a bespoke $\text{Voice}_{\text{pass}}$, we follow Wood's

(2015) analysis (see seminal ideas in Embick 2004; see also Schäfer 2008:174ff, Bruening 2013, and Kastner 2020:171ff), in which the passive participle contains a normal Voice head whose variable over agents is existentially bound, not by a DP, but by an Asp head (Pass in Bruening 2013 and Kastner 2020), realized via the participial morphology, that is, *-t* in (i).

[9]Our analysis comes close to Šereikaitė's (2018), where the LP is also merged directly to the combination of v and the verbal root. Šereikaitė argues extensively that the LP in Lithuanian cannot be phrasal. However, if it is not phrasal, it should be a head, but seemingly not a projecting one, as it is v that projects. Beyond this phrase-structural issue, it is unclear to us how Šereikaitė's analysis captures the predicative and telicity-related nature of the LP. See also section 7.2 for a phonological motivation for the phrasal status of LPs (as ResPs) in prefixed verbs.

[10]This is motivated by the multiple scope-taking possibilities of these prefixes. Thus, as described by Arkadiev (2012), restrictive *te-* 'only' may take scope over the subject, the entire VP, the object, or the indirect object, as shown in his example in (i). Prefixes of this kind are typologically rare, but Arkadiev (2012) observes that they exhibit similarities with verbal prefixes in Mayali, as described by Evans (1995), who argues that they are A-quantifiers.

(i) Kaz-ys te-dovanoj-o mergait-ėms knyg-as.
Kaz-NOM.SG *te*-give-PST.3 girl-DAT.PL book-ACC.PL
i. 'Only Kazys gave books to girls.'
ii. 'Kazys only gave books to girls <and not sold flowers to boys>.'
iii. 'Kazys gave books only to girls <and not to boys>.'
iv. 'Kazys gave girls only books <and not flowers>.'
v. 'Kazys only gave books to girls <and not sold books to girls>.'

While *be-* can be used as continuative aspect, it has several other aspectual uses, as well as modal and evaluative uses (progressive, avertive, admirative, dubitative, attenuative, universal; see Arkadiev 2011, 2012.) Arkadiev (2012), from a typological perspective, stresses that the aspectual uses of *be-* can be traced back to a common historical origin of its use as the continuative, but establishing a relationship between its modal-evaluative uses, historically or synchronically, is difficult. While both *be-* and *te-* have several different uses, *be-* also has at least one A-quantificational use: in "attenuative" constructions, *be-* behaves like the restrictive *te-* (the two being interchangeable). This is observed by Arkadiev (2012) with the example in (ii), where the two sentences with *te-* and *be-* are synonymous. Furthermore, both options can have two interpretations: the restriction can associate either with *antradieniais* 'on Tuesdays' or with *į universitetą* 'to university', depending on focus, similarly to what we saw in (i).

(ii) Jis antradieniais te/be-nu-ei-dav-o į universitetą.
he.NOM Tuesday.INSTR.PL *te*/*be*-LP-go-HAB-PST.3 to university.ACC
(i) He came to university only on Tuesdays.
(ii) On Tuesdays, he only came to university. (Adapted from Arkadiev 2012)

Korostenskienė (2022) connects *te-* and *be-* to ForceP, in the CP domain. This would still be consistent with our proposal for linearization: *te-* and *be-* will be linearized to the left of the verbal base and the lexical prefix, as long as *te-* and *be-* are specifiers and thus form an independent @-span. That this is the case is suggested by prosodic evidence: these prefixes can be excorporated, that is, they can form a prosodic domain independently of the verbal one in contexts of question-answer pairs and ellipsis.

(iii) A: Te-valgo jis saldainį?
te-eat.3 he.NOM candy.ACC
'Let him eat candy?'

B: Te.
te
'Yes, let him.'

(iv) Jis te-be-valgė saldainį, o aš ne-be.
he.NOM *te*-PROG-eat.3 candy.ACC and I.NOM NEG-PROG
'He was still eating the candy, and I wasn't anymore.'

Negative *ne-* also shows phrasal behavior. It behaves like other superlexical prefixes in that it can form an independent prosodic domain in excorporation, and can be modified (e.g. by 'certainly'), which suggests that it is phrasal.

(v) A: Ar ne-nori eiti pas gydytoją?
Q NEG-want.PRS.2SG go.INF to doctor.ACC
'You don't want to go to the doctor's?'

B: Tikrai ne.
certainly NEG
'Certainly not.'

(vi) Aš norėjau išeit pabėgioti, o ji ne.
I.NOM want.PST.1SG go.INF run.INF and she NEG
'I wanted to go for a run, but she didn't.'

Further evidence that *ne-* corresponds to an adverbial phrase licensing a Neg projection in the verbal extended projection, rather than to the head Neg itself, is that it can be attached to non-eventive nominals, as in *ne-draugas* 'non-friend', 'enemy'.

Another curious fact about *te-* and *ne-* is that they seem to be in complementary distribution: Arkadiev (2012:8) reports that examples with both are "extremely rare and considered to be 'borderline'." If so, they might be connected to a Polarity projection, with *ne-* being a specifier of a negative polarity head, and *te-* of a positive polarity head. Furthermore, if they are truly in complementary distribution, then if *te-* is phrasal, as has been

argued, it follows that *ne-* has to be phrasal as well.

The full analysis of *te-* and *be-* is beyond the scope of this paper, since there are many puzzling facts about these prefixes, both semantic (the abundance of interpretations) and syntactic-morphological (they have cooccurrence restrictions and *be* seems to prefer participles). Our main point here is about the relative position of these elements in the verb, and we see from the morphology that *be-* is above the verb but below *te-* and *ne-*. Even if *be-* is located within a continuative Asp projection, as an anonymous reviewer suggests, the final product of linearization with respect to *si* will end up identical—Asp, being situated above Voice but below CP, will be linearized to the right of *te-* but to the left of *si*.

[11]Evidence that this movement is syntactic and, indeed, phrasal, is provided by cases where different phrases, like the subject, negation, certain adverbs, and so on, intervene between the clitic and the inflected verb of which the clitic is an argument (Barbosa 1996:7).

[12]While the clitic ends up tucked at the right edge of a prosodic word to avoid the constraint, Barbosa (1996:35) in fact proposes a tentative syntactic explanation to account for the displacement, rather than an operation like Prosodic Inversion (Halpern 1992).

[13]Other types of account of similar phenomena, not relying on syntactic movement, prosodic rules, or OT constraints, include prosodic templates (e.g., Hock 1996 for Vedic, among other languages) and prosodic subcategorization frames (e.g., Chung 2003 for weak pronouns in Chamorro and Herd 2003 for adverb particles in Maori).

[14]For another OT-account of a prosody-sensitive reordering effect at the sentential level see Bennett et al.'s (2016) treatment of weak object pronoun postposing in Irish.

[15]We provide informal descriptions and refer the reader to Bye and Svenonius 2012 for the full technical ones.

[16]LINEARITY, as used by, for instance, in Bye and Svenonius 2012, counts the precise number of instances of a particular segment of the output reversing precedence relations

with segments of the input. In the tableaux below, however, we abstract away from that information, as it is not relevant for the evaluation.

[17]We state here that, before his article was published, we were unaware of Kushnir's research on *si*. The main sources of inspiration for our analysis are Barbosa's (1996) approach to the position of pronominal clitics in EP, Acedo-Matellán and Kwapiszewski's (2026) Spanning analysis of prefixed predicates in weak satellite-framed languages, and Bye and Svenonius's (2012) treatment of antitropal exponents.

[18]In fact, the second position is epiphenomenal in our account. It may even not be literally second position, as with prefix stacks—see Šereikaitė 2017. What we predict, rightfully, is that one prosodic constituent to the left of *si* is enough to produce a well-formed sequence. See also Arregi and Nevins 2012 for the idea that Lithuanian *si* represents a second position phenomenon within the word. Their account, however, is not prosodic and hence not fully descriptively adequate (see section 2).

[19]We emphasize that the account of the diphthong vs monophthong difference exemplified here, and the accompanying correlation with the absence vs presence of event- and argument-structure properties rests on the assumption that the LP of verbs (and eventive nominalizations) ultimately corresponds to a phrase introducing eventuality semantics and sustaining arguments (ResP), while the one in non-eventive nominals like *nuo-lauža* is directly merged with the root. See also Svenonius 2008, where it is observed that Russian LPs in verbs also form phonological domains of their own and proposes that this is related to their phrasal status. See Acedo-Matellán and Kwapiszewski 2026 for similar observations based on the morphophonology of resultative prefixes in Latin and Ancient Greek.

[20]To the best of our knowledge, the *-si*/*-s* alternation has not been dealt with in the literature. It is unclear whether it is morphologically or phonologically conditioned. In favor of the former explanation, *-si* is disallowed in (a) while it is optional in (b), even though

both forms end in exactly the same vowel.

(i) a. mat-ot[e:]-s(*i)
see-2.PL.PRES-*si*
'you are visible'

b. mat-[e:]-s(i)
see-3.PAST-*si*
'was visible'

Still a phonological explanation could be pursued in terms of stress assignment and number of moras on the basis of the contrasts such as (i), as both cases are stressed on the first mora but they show a different number of moras. It is also telling that a similar alternation as in (b) is attested in infinitives: *matyt*(*i*) 'to see'. We leave this issue for further research.

[21]This way, postposed *si* behaves as other run-of-the-mill affixes, such as the future affix *-s*, which are integrated within the prosodic word. Indeed, future *-s* also prevents diphthongs from clipping, as in the third person forms ***lie****-s* 'will pour' (cf. **li-s*) and *dain****uo****-s* 'will sing' (cf. **dainu-s*)

[22]In fact, here the first sibilant anticipatorily assimilates to the /s/ of the reflexive.

[23]Similarly, Korostenskienė (2014) describes *si* as attached to the leftmost affix, although linearization (suffixation/prefixation) is not discussed explicitly.